

Computational Linear Algebra: PCA Homework

Exercise 0.0: Initialization

Fill the missing values in this text box. **Remove the information of the second student if you are not working in a team.**

Academic Year: 2025/2026

Team Members (Alphabetical Order):

1. Miori, Arianna (s353777);

Now, fill the python list below with the Student IDs of the team

```
StudentIDs = [353777] # <----- Fill this list with Student IDs of the team
```

Exercise 0.1: Starting Code-Cell

Download the .csv files from the web page of the course (*responses_hw.csv* and *columns_hw.csv*) and past them in **the same folder of this notebook**.

Then, run the cell below, **without modifying any line of code**.

The output of this code cell is **your personal subset of the original dataset**, with 2/3 of the original features (i.e., columns) and 3/4 of the original persons (i.e., rows).

ATTENTION: DO NOT CHANGE THE CODE INSIDE THE FOLLOWING CELL, ANY CHANGE CAN INVALIDATE THE HOMEWORK!

```
#####
##### DO NOT CHANGE THE CODE IN THIS CELL #####
#####

import numpy as np
import pandas as pd
from IPython.display import display

var_entertainment_feat_types = ['Interests', 'Movies', 'Music']
var_personal_feat_types = ['Finance', 'Phobias']
fixed_feat_types = ['Personality', 'Health']

label_types = ['Demographic']

variables_by_type = {
    'Demographics': ['Age', 'Height', 'Weight', 'Number of siblings',
                     'Gender', 'Hand', 'Education', 'Only child',
```

```

'Home Town Type',
    'Home Type'],
'Finance': ['Finances', 'Shopping centres', 'Branded clothing',
    'Entertainment spending', 'Spending on looks',
    'Spending on gadgets', 'Spending on healthy eating'],
'Health': ['Smoking', 'Alcohol', 'Healthy eating'],
'Interests': ['History', 'Psychology', 'Politics', 'Mathematics',
    'Physics', 'Internet', 'PC', 'Economy Management',
    'Biology', 'Chemistry', 'Reading', 'Geography',
    'Foreign languages', 'Medicine', 'Law', 'Cars',
    'Art exhibitions', 'Religion', 'Countryside,
outdoors',
    'Dancing', 'Musical instruments', 'Writing',
'Passive sport',
    'Active sport', 'Gardening', 'Celebrities',
'Shopping',
    'Science and technology', 'Theatre', 'Fun with
friends',
    'Adrenaline sports', 'Pets'],
'Movies': ['Movies', 'Horror', 'Thriller', 'Comedy', 'Romantic',
    'Sci-fi', 'War', 'Fantasy/Fairy tales', 'Animated',
    'Documentary', 'Western', 'Action'],
'Music': ['Music', 'Slow songs or fast songs', 'Dance', 'Folk',
    'Country', 'Classical music', 'Musical', 'Pop', 'Rock',
    'Metal or Hardrock', 'Punk', 'Hiphop, Rap', 'Reggae,
Ska',
    'Swing, Jazz', 'Rock n roll', 'Alternative', 'Latino',
    'Techno, Trance', 'Opera'],
'Personality': ['Daily events', 'Prioritising workload',
    'Writing notes', 'Workaholism', 'Thinking ahead',
    'Final judgement', 'Reliability', 'Keeping
promises',
    'Loss of interest', 'Friends versus money',
'Funniness',
    'Fake', 'Criminal damage', 'Decision making',
'Elections',
    'Self-criticism', 'Judgment calls',
'Hypochondria',
    'Empathy', 'Eating to survive', 'Giving',
    'Compassion to animals', 'Borrowed stuff',
    'Loneliness', 'Cheating in school', 'Health',
    'Changing the past', 'God', 'Dreams', 'Charity',
    'Number of friends', 'Punctuality', 'Lying',
'Waiting',
    'New environment', 'Mood swings', 'Appearance and
gestures',
    'Socializing', 'Achievements', 'Responding to a
serious letter',
    'Children', 'Assertiveness', 'Getting angry',

```

```

        'Knowing the right people', 'Public speaking',
        'Unpopularity', 'Life struggles', 'Happiness in
life',
        'Energy levels', 'Small - big dogs',
'Personality',
        'Finding lost valuables', 'Getting up', 'Interests
or hobbies',
        "Parents' advice", 'Questionnaires or polls',
'Internet usage'],
        'Phobias': ['Flying', 'Storm', 'Darkness', 'Heights', 'Spiders',
'Snakes',
        'Rats', 'Ageing', 'Dangerous dogs', 'Fear of public
speaking']
    }

labels = variables_by_type['Demographics']
features_all = []
for tt in variables_by_type.keys():
    if tt != 'Demographics':
        features_all += variables_by_type[tt]

def which_features(*StudentIDs):
    random_seed = min(StudentIDs)
    np.random.seed(random_seed)
    features_ = np.random.choice(features_all, int((2 *
len(features_all)) / 3), replace=False).tolist()
    features = []
    features_by_type = {tt: [] for tt in variables_by_type.keys() if
tt != 'Demographics'}
    for tt in variables_by_type.keys():
        ft_list = variables_by_type[tt]
        for ii in range(len(ft_list)):
            if ft_list[ii] in features_:
                features.append(ft_list[ii])
                features_by_type[tt].append(ft_list[ii])

    return features, features_by_type

features, features_by_type = which_features(*StudentIDs)

print(f'*** THESE ARE THE {len(features)} SELECTED FEATURES (SEE
VARIABLE features):')
for ff in features:
    print(f'{ff}')
print('*****')
print('')
print('*** SELECTED FEATURES BY TYPES (SEE VARIABLE
features_by_type):')
for tt in features_by_type.keys():
    print(f'{tt}: {features_by_type[tt]}')

```

```

    print('')
print('*****')
print('')
print('*** THESE ARE THE LABELS (SEE VARIABLE labels):')
for ll in labels:
    print(f'{ll}')
print('*****')

def which_rows(df, frac, *StudentIDs):
    random_seed = min(StudentIDs)
    df_ = df.sample(frac=frac, random_state=random_seed)
    return df_

responses_hw = pd.read_csv('responses_hw.csv', index_col=0)
responses = which_rows(responses_hw, 0.75, *StudentIDs)
responses = responses.loc[:, features + labels]

responses_ft = responses.loc[:, features]
responses_lb = responses.loc[:, labels]

print('')
print('*** THIS IS YOUR PERSONAL DATASET (features AND labels TOGETHER, SEE VARIABLE responses)')
display(responses)
print('')
print('*** THIS IS YOUR PERSONAL DATASET (features, SEE VARIABLE responses_ft)')
display(responses_ft)
print('')
print('*** THIS IS YOUR PERSONAL DATASET (labels, SEE VARIABLE responses_lb)')
display(responses_lb)

random_seed = min(StudentIDs)
np.random.seed(random_seed)

your_scaler = np.random.choice(['StandardScaler', 'MinMaxScaler'])

*** THESE ARE THE 93 SELECTED FEATURES (SEE VARIABLE features):
Finances
Shopping centres
Entertainment spending
Spending on looks
Spending on gadgets
Spending on healthy eating
Smoking
Alcohol
Psychology
Politics
Internet

```

PC
Economy Management
Reading
Geography
Law
Countryside, outdoors
Dancing
Writing
Gardening
Celebrities
Shopping
Science and technology
Theatre
Adrenaline sports
Pets
Movies
Horror
Thriller
Comedy
Animated
Western
Action
Music
Dance
Folk
Classical music
Musical
Rock
Metal or Hardrock
Punk
Hiphop, Rap
Reggae, Ska
Swing, Jazz
Rock n roll
Alternative
Latino
Prioritising workload
Thinking ahead
Final judgement
Reliability
Friends versus money
Funniness
Fake
Criminal damage
Self-criticism
Judgment calls
Hypochondria
Eating to survive
Giving

Borrowed stuff
Loneliness
Cheating in school
Health
Changing the past
God
Charity
Number of friends
Waiting
New environment
Mood swings
Appearance and gestures
Achievements
Assertiveness
Getting angry
Unpopularity
Life struggles
Happiness in life
Energy levels
Small - big dogs
Personality
Finding lost valuables
Getting up
Interests or hobbies
Parents' advice
Questionnaires or polls
Internet usage
Flying
Darkness
Heights
Snakes
Rats
Dangerous dogs

*** SELECTED FEATURES BY TYPES (SEE VARIABLE features_by_type):

Finance: ['Finances', 'Shopping centres', 'Entertainment spending',
'Spending on looks', 'Spending on gadgets', 'Spending on healthy
eating']

Health: ['Smoking', 'Alcohol']

Interests: ['Psychology', 'Politics', 'Internet', 'PC', 'Economy
Management', 'Reading', 'Geography', 'Law', 'Countryside, outdoors',
'Dancing', 'Writing', 'Gardening', 'Celebrities', 'Shopping', 'Science
and technology', 'Theatre', 'Adrenaline sports', 'Pets']

Movies: ['Movies', 'Horror', 'Thriller', 'Comedy', 'Animated',
'Western', 'Action']

Music: ['Music', 'Dance', 'Folk', 'Classical music', 'Musical', 'Rock', 'Metal or Hardrock', 'Punk', 'Hiphop, Rap', 'Reggae, Ska', 'Swing, Jazz', 'Rock n roll', 'Alternative', 'Latino']

Personality: ['Prioritising workload', 'Thinking ahead', 'Final judgement', 'Reliability', 'Friends versus money', 'Funniness', 'Fake', 'Criminal damage', 'Self-criticism', 'Judgment calls', 'Hypochondria', 'Eating to survive', 'Giving', 'Borrowed stuff', 'Loneliness', 'Cheating in school', 'Health', 'Changing the past', 'God', 'Charity', 'Number of friends', 'Waiting', 'New environment', 'Mood swings', 'Appearance and gestures', 'Achievements', 'Assertiveness', 'Getting angry', 'Unpopularity', 'Life struggles', 'Happiness in life', 'Energy levels', 'Small - big dogs', 'Personality', 'Finding lost valuables', 'Getting up', 'Interests or hobbies', 'Parents' advice', 'Questionnaires or polls', 'Internet usage']

Phobias: ['Flying', 'Darkness', 'Heights', 'Snakes', 'Rats', 'Dangerous dogs']

*** THESE ARE THE LABELS (SEE VARIABLE labels):

Age
Height
Weight
Number of siblings
Gender
Hand
Education
Only child
Home Town Type
Home Type

*** THIS IS YOUR PERSONAL DATASET (features AND labels TOGETHER, SEE VARIABLE responses)

	Finances	Shopping centres	Entertainment spending	Spending on looks \
316	2	4	4	
5				
53	3	3	3	
2				
568	1	5	5	
5				
549	2	5	5	
5				
295	3	5	3	
3				

child \	Number of siblings	Gender	Hand	Education	Only
316	6	female	right	secondary school	
no					
53	0	female	right	college/bachelor degree	
yes					
568	0	male	right	secondary school	
yes					
549	0	male	right	secondary school	
yes					
295	2	male	right	college/bachelor degree	
no					
..	...	...	...	...	...
...					
419	1	female	right	secondary school	
no					
541	2	female	right	secondary school	
no					
47	1	male	right	college/bachelor degree	
no					
204	2	male	right	secondary school	
no					
628	5	female	left	primary school	
no					

	Home Town Type	Home Type
316	city	block of flats
53	village	house/bungalow
568	city	block of flats
549	city	block of flats
295	city	block of flats
..	...	...
419	village	block of flats
541	city	block of flats
47	city	block of flats
204	city	block of flats
628	city	block of flats

[506 rows x 103 columns]

*** THIS IS YOUR PERSONAL DATASET (features, SEE VARIABLE responses_ft)

looks \	Finances	Shopping centres	Entertainment spending	Spending on
316	2	4	4	
5				
53	3	3	3	
2				

568	1	5	5
5			
549	2	5	5
5			
295	3	5	3
3			
..	...	...	...
...			
419	3	3	3
3			
541	3	1	2
3			
47	5	1	3
1			
204	5	1	3
3			
628	3	5	2
3			
Spending on gadgets		Spending on healthy eating	
Smoking \			
316	3	3	former smoker
53	1	3	never smoked
568	5	2	tried smoking
549	2	4	never smoked
295	4	4	tried smoking
..	...	...	...
419	2	2	tried smoking
541	3	4	current smoker
47	1	4	tried smoking
204	4	3	never smoked
628	2	4	former smoker
Alcohol		Psychology	Politics ... Interests or
hobbies \			
316	drink a lot	2	3 ... 4
53	never	3	3 ... 4
568	social drinker	3	4 ... 5

549	drink a lot	3	5	...	5
295	social drinker	5	3	...	5
..	...	...	...	...	...
419	social drinker	3	1	...	5
541	social drinker	4	5	...	1
47	social drinker	5	4	...	5
204	social drinker	1	1	...	5
628	social drinker	2	2	...	4
	Parents' advice	Questionnaires or polls		Internet usage	Flying
316	3		1	few hours a day	2
53	2		3	few hours a day	1
568	2		1	few hours a day	1
549	2		1	most of the day	1
295	3		4	few hours a day	1
..	...		...	...	...
419	3		3	few hours a day	3
541	3		1	few hours a day	1
47	3		2	most of the day	1
204	3		1	most of the day	1
628	3		3	few hours a day	3
	Darkness	Heights	Snakes	Rats	Dangerous dogs
316	2	2	4	3	3
53	2	4	3	2	5
568	1	3	1	1	1
549	1	2	5	1	4
295	3	4	4	4	4
..	...	...	...	...	...
419	4	3	5	4	5

541	1	2	1	5	3
47	1	2	3	2	3
204	1	4	3	1	3
628	3	4	5	5	5

[506 rows x 93 columns]

*** THIS IS YOUR PERSONAL DATASET (labels, SEE VARIABLE responses_lb)

	Age	Height	Weight	Number of siblings	Gender	Hand \
316	18	163	58	6	female	right
53	24	160	58	0	female	right
568	19	187	88	0	male	right
549	18	176	76	0	male	right
295	18	167	56	2	male	right
..	...	...	...	...	...	...
419	19	164	55	1	female	right
541	18	160	55	2	female	right
47	23	175	72	1	male	right
204	20	178	68	2	male	right
628	18	165	45	5	female	left

	Education	Only child	Home	Town	Type	Home Type
316	secondary school	no		city		block of flats
53	college/bachelor degree	yes		village		house/bungalow
568	secondary school	yes		city		block of flats
549	secondary school	yes		city		block of flats
295	college/bachelor degree	no		city		block of flats
..	...	...		...		...
419	secondary school	no		village		block of flats
541	secondary school	no		city		block of flats
47	college/bachelor degree	no		city		block of flats
204	secondary school	no		city		block of flats
628	primary school	no		city		block of flats

[506 rows x 10 columns]

Exercise 0.2: Importing Modules

In the following cell, import all the modules you think are necessary for doing the homework, **among the ones listed and used during the laboratories of the course**.

For reproducibility, **no extra modules are allowed**.

DO NOT IMPORT NUMPY NOR PANDAS, they are already imported.

```
# DO NOT IMPORT NUMPY AND PANDAS - Already imported
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import OrdinalEncoder
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, silhouette_samples
```

Exercise 1. Dataset Preprocessing

In this exercise, you have to do the following operations on the dataset of the features **responses_ft**:

1. Create a new dataframe called **responses_ft_enc** by encoding the categorical features (if they exist), motivating your choices;
2. Create a new dataframe called **responses_ft_pp** by preprocessing the data in **responses_ft_enc**, according to the scaler reported in the cell below.

```
print(f'*** YOU HAVE TO APPLY A PREPROCESSING USING THE  
{your_scaler}')  
  
*** YOU HAVE TO APPLY A PREPROCESSING USING THE MinMaxScaler
```

For doing this exercise, fill the cells below following the instructions you read.

Describe and motivate the encoding operations you will apply (max 150 words):

The dataset contains only three categorical features that require encoding. Since the responses of these features follow an incremental order, it is appropriate to use an Ordinal Encoder. In particular, this encoder works well for the "Alcohol" and "Internet usage" categories in which there is a clear increasing order in the responses. For the "Smoking" category, the responses ("never smoked", "tried smoking", "former smoker", "current smoker") are ordinal, but they do not follow a strictly linear progression and, in general, a OneHotEncoder may be more appropriate. However, using an Ordinal Encoder is acceptable in this context, as PCA will be applied; this approach ensures that all features maintain a consistent ordered structure, which supports both the performance and interpretability of PCA.

Describe the preprocessing operation you will apply and comment the effects it may have on the data (max 150 words):

The MinMaxScaler transforms features by rescaling each feature to a given range. In particular, each feature is rescaled using the minimum and the maximum values and mapped within the range [0,1]. Although in this case it is required to use the MinMaxScaler, it is appropriate here because it preserves the ordered structure of the dataset.

Write the code for performing the encoding and preprocessing operations of the exercise. Show the encoded data and the preprocessed data you obtain, plus any additional table/value that can be useful for commenting the results:

```
# I check which are the columns that have object as type
print(responses_ft.select_dtypes(include='object').columns)

objs = ["Smoking", "Alcohol", "Internet usage"]

# I divide the dataframe in the categorical and numerical values
obj_responses_ft = responses_ft[objs]
int_responses_ft = responses_ft.drop(columns=objs)

# Read the possible value in a specific categorical attribute
for col in objs:
    print(obj_responses_ft[col].value_counts())

categories = [["never smoked", "tried smoking", "former smoker",
"current smoker"],
              ["never", "social drinker", "drink a lot"],
              ["less than an hour a day", "few hours a day", "most of
the day"]]

# Encode the categorical data using OrdinalEncoder
oe = OrdinalEncoder(categories=categories)
obj0E = oe.fit_transform(obj_responses_ft[objs])

obj_responses_ft_enc= pd.DataFrame(
    obj0E,
    columns = objs,
    index = obj_responses_ft.index
)

# Generate the encoded dataframe by concatenate the numerical and
encoded categorical values
responses_ft_enc = pd.concat([int_responses_ft, obj_responses_ft_enc],
axis=1)
display(responses_ft_enc)

# Apply the MinMax scaler to the dataframe
minmax_scaler = MinMaxScaler()

responses_ft_pp = pd.DataFrame(
```

```

minmax_scaler.fit_transform(responses_ft_enc),
columns=responses_ft_enc.columns,
index=responses_ft_enc.index
)

display(responses_ft_pp)
Index(['Smoking', 'Alcohol', 'Internet usage'], dtype='object')
Smoking
tried smoking      224
never smoked       114
former smoker       92
current smoker      76
Name: count, dtype: int64
Alcohol
social drinker     339
drink a lot        115
never              52
Name: count, dtype: int64
Internet usage
few hours a day    383
less than an hour a day  68
most of the day    55
Name: count, dtype: int64

```

	Finances	Shopping centres	Entertainment spending	Spending on
looks \				
316	2	4	4	
5				
53	3	3	3	
2				
568	1	5	5	
5				
549	2	5	5	
5				
295	3	5	3	
3				
...	...	...	...	
...				
419	3	3	3	
3				
541	3	1	2	
3				
47	5	1	3	
1				
204	5	1	3	
3				
628	3	5	2	
3				

Spending on gadgets			Spending on healthy eating			Psychology		
Politics \								
316		3			3			2
3								
53		1			3			3
3								
568		5			2			3
4								
549		2			4			3
5								
295		4			4			5
3								
..		...			...			...
...								
419		2			2			3
1								
541		3			4			4
5								
47		1			4			5
4								
204		4			3			1
1								
628		2			4			2
2								
Internet	PC	...	Questionnaires or polls			Flying	Darkness	
Heights \								
316	3	4	...		1	2		2
2								
53	3	1	...		3	1		2
4								
568	4	4	...		1	1		1
3								
549	4	5	...		1	1		1
2								
295	5	4	...		4	1		3
4								
..	...	..	...		...	...		...
...								
419	4	1	...		3	3		4
3								
541	5	2	...		1	1		1
2								
47	5	4	...		2	1		1
2								
204	5	4	...		1	1		1
4								
628	5	4	...		3	3		3
4								

	Snakes	Rats	Dangerous dogs	Smoking	Alcohol	Internet usage
316	4	3	3	2.0	2.0	1.0
53	3	2	5	0.0	0.0	1.0
568	1	1	1	1.0	1.0	1.0
549	5	1	4	0.0	2.0	2.0
295	4	4	4	1.0	1.0	1.0
..	...	...	...	...	...	...
419	5	4	5	1.0	1.0	1.0
541	1	5	3	3.0	1.0	1.0
47	3	2	3	1.0	1.0	2.0
204	3	1	3	0.0	1.0	2.0
628	5	5	5	2.0	1.0	1.0

[506 rows x 93 columns]

	Finances	Shopping centres	Entertainment spending	Spending on looks \
316	0.25	0.75	0.75	1.00
53	0.50	0.50	0.50	0.25
568	0.00	1.00	1.00	1.00
549	0.25	1.00	1.00	1.00
295	0.50	1.00	0.50	0.50
..	...	...	...	...
419	0.50	0.50	0.50	0.50
541	0.50	0.00	0.25	0.50
47	1.00	0.00	0.50	0.00
204	1.00	0.00	0.50	0.50
628	0.50	1.00	0.25	0.50

	Spending on gadgets	Spending on healthy eating	Psychology
316	0.50	0.50	0.25
53	0.00	0.50	0.50
568	1.00	0.25	0.50
549	0.25	0.75	0.50

1.00			
295	0.75	0.75	1.00
0.50			
..	...	...	...
...			
419	0.25	0.25	0.50
0.00			
541	0.50	0.75	0.75
1.00			
47	0.00	0.75	1.00
0.75			
204	0.75	0.50	0.00
0.00			
628	0.25	0.75	0.25
0.25			

	Internet	PC	...	Questionnaires or polls	Flying	Darkness
Heights \						
316	0.50	0.75	...	0.00	0.25	0.25
0.25						
53	0.50	0.00	...	0.50	0.00	0.25
0.75						
568	0.75	0.75	...	0.00	0.00	0.00
0.50						
549	0.75	1.00	...	0.00	0.00	0.00
0.25						
295	1.00	0.75	...	0.75	0.00	0.50
0.75						
..	...	...	...	...	...	...
...						
419	0.75	0.00	...	0.50	0.50	0.75
0.50						
541	1.00	0.25	...	0.00	0.00	0.00
0.25						
47	1.00	0.75	...	0.25	0.00	0.00
0.25						
204	1.00	0.75	...	0.00	0.00	0.00
0.75						
628	1.00	0.75	...	0.50	0.50	0.50
0.75						

	Snakes	Rats	Dangerous	dogs	Smoking	Alcohol	Internet usage
316	0.75	0.50		0.50	0.666667	1.0	0.5
53	0.50	0.25		1.00	0.000000	0.0	0.5
568	0.00	0.00		0.00	0.333333	0.5	0.5
549	1.00	0.00		0.75	0.000000	1.0	1.0
295	0.75	0.75		0.75	0.333333	0.5	0.5
..	...	...		...	...	...	...
419	1.00	0.75		1.00	0.333333	0.5	0.5

541	0.00	1.00	0.50	1.000000	0.5	0.5
47	0.50	0.25	0.50	0.333333	0.5	1.0
204	0.50	0.00	0.50	0.000000	0.5	1.0
628	1.00	1.00	1.00	0.666667	0.5	0.5

[506 rows x 93 columns]

Comment the results obtained after the preprocessing operation (max 100 words):

As expected, the categorical features have become numerical and after the preprocessing all the values are between 0 and 1. The relative ordering of the features has been preserved, ensuring that the dataset maintains its structure. They can now be used for variance analysis and for PCA.

Exercise 2. Analyzing the Variance and the PCs

In this exercise, you have to do the following operations:

1. compute and visualize the variance of all the features in *responses_ft_enc* and *responses_ft_pp*;
2. compute all the n Principal Components (PCs) for *responses_ft_enc* and *responses_ft_pp*, separately, and visualize the curves of the cumulative explained variances.

For doing this exercise, fill the cells below following the instructions you read.

Write the code for computing and visualizing the variance of the features of the two datasets:

```
# For adding color to each category
Finance = ['Finances', 'Shopping centres', 'Entertainment spending',
'Spending on looks', 'Spending on gadgets', 'Spending on healthy
eating']
Health = ['Smoking', 'Alcohol']
Interests = ['Psychology', 'Politics', 'Internet', 'PC', 'Economy
Management', 'Reading', 'Geography', 'Law', 'Countryside, outdoors',
'Dancing', 'Writing', 'Gardening', 'Celebrities', 'Shopping', 'Science
and technology', 'Theatre', 'Adrenaline sports', 'Pets']
Movies = ['Movies', 'Horror', 'Thriller', 'Comedy', 'Animated',
'Western', 'Action']
Music = ['Music', 'Dance', 'Folk', 'Classical music', 'Musical',
'Rock', 'Metal or Hardrock', 'Punk', 'Hiphop, Rap', 'Reggae, Ska',
'Swing, Jazz', 'Rock n roll', 'Alternative', 'Latino']
Personality = ['Prioritising workload', 'Thinking ahead', 'Final
judgement', 'Reliability', 'Friends versus money', 'Funniness',
'Fake', 'Criminal damage', 'Self-criticism', 'Judgment calls',
'Hypochondria', 'Eating to survive', 'Giving', 'Borrowed stuff',
'Loneliness', 'Cheating in school', 'Health', 'Changing the past',
'God', 'Charity', 'Number of friends', 'Waiting', 'New environment',
'Mood swings', 'Appearance and gestures', 'Achievements',
'Assertiveness', 'Getting angry', 'Unpopularity', 'Life struggles',
```

```

'Happiness in life', 'Energy levels', 'Small - big dogs',
'Personality', 'Finding lost valuables', 'Getting up', 'Interests or
hobbies', "Parents' advice", 'Questionnaires or polls', 'Internet
usage']
Phobias = ['Flying', 'Darkness', 'Heights', 'Snakes', 'Rats',
'Dangerous dogs']

categories = {
    'Finance': Finance,
    'Health': Health,
    'Interests': Interests,
    'Movies': Movies,
    'Music': Music,
    'Personality': Personality,
    'Phobias': Phobias
}

cmap = plt.get_cmap('Set3')

category_colors = {
    'Finance': cmap(0),
    'Health': cmap(1),
    'Interests': cmap(2),
    'Movies': cmap(3),
    'Music': cmap(4),
    'Personality': cmap(5),
    'Phobias': cmap(6)
}

column_color = {}
for cat, cols in categories.items():
    for col in cols:
        column_color[col] = category_colors[cat]

# For dividing the two groups of categories
entertainment = ['Interests', 'Movies', 'Music']
personality = ['Phobias', 'Health', 'Personality', 'Finance']

ordered_groups = entertainment + personality
ordered_columns = []
for group in ordered_groups:
    ordered_columns.extend(categories[group])

responses_ft_enc = responses_ft_enc[ordered_columns]
responses_ft_pp = responses_ft_pp[ordered_columns]

# compute the variance
var_responses_ft_enc = responses_ft_enc.var(axis=0)
var_responses_ft_pp = responses_ft_pp.var(axis=0)

```

```

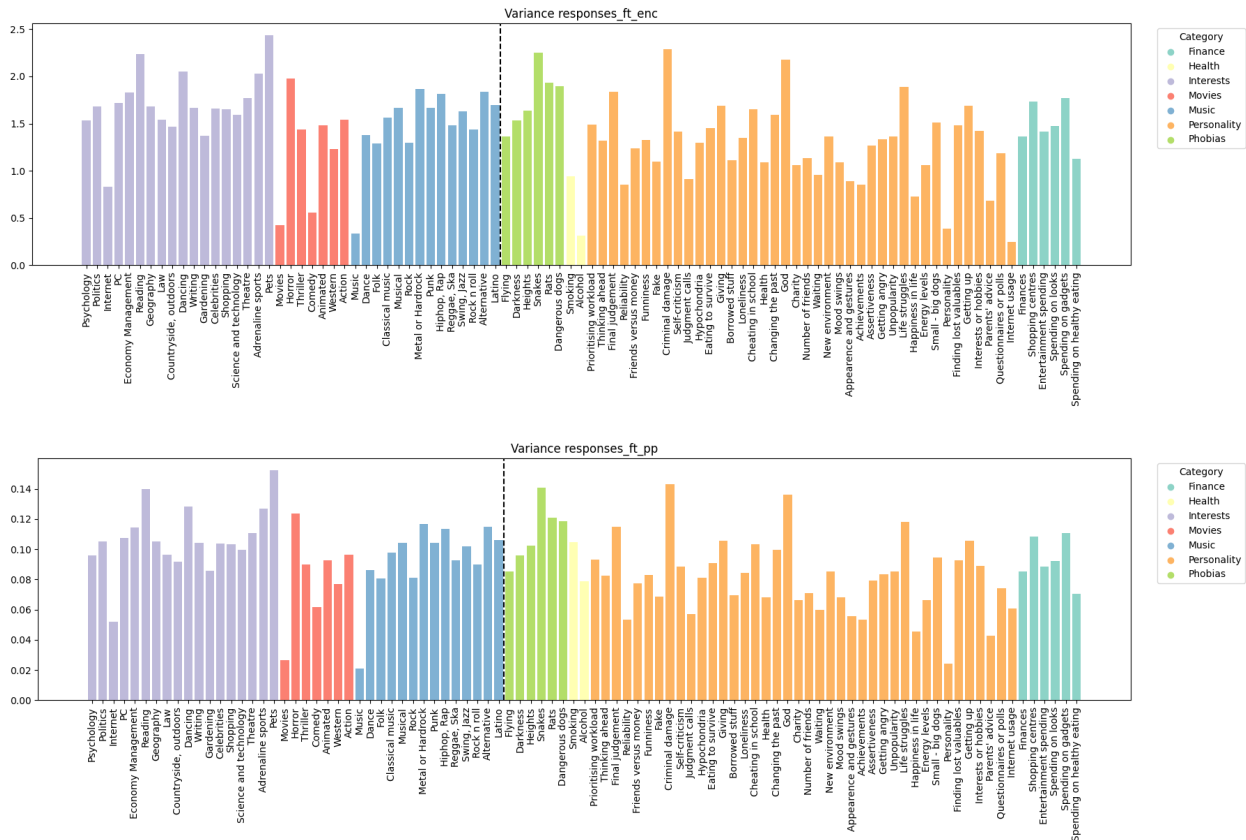
# for the color in the graph
colors_enc = [column_color.get(col, 'gray') for col in
var_responses_ft_enc.index]
colors_pp = [column_color.get(col, 'gray') for col in
var_responses_ft_pp.index]

# for adding a vline in the plot to separate the two groups
entertainment_cols = []
for g in ['Interests', 'Movies', 'Music']:
    entertainment_cols.extend(categories[g])
vline_pos = len(entertainment_cols) - 0.5

# plotting the variance of the encoded dataset
plt.figure(figsize=(18, 6))
plt.bar(np.arange(len(var_responses_ft_enc)), var_responses_ft_enc,
color = colors_enc)
plt.axvline(x=vline_pos, color='black', linestyle='--', linewidth=1.5)
for cat, color in category_colors.items():
    plt.scatter([], [], color=color, label=cat)
plt.legend(title='Category', bbox_to_anchor=(1.02,1), loc='upper
left')
plt.xticks(np.arange(len(var_responses_ft_enc)),
            var_responses_ft_enc.index,
            rotation=90,
            fontsize=10)
plt.title("Variance responses_ft_enc")
plt.tight_layout()
plt.show()

# plotting the variance of the preprocessed dataset
plt.figure(figsize=(18, 6))
plt.bar(np.arange(len(var_responses_ft_pp)), var_responses_ft_pp,
color = colors_pp)
plt.axvline(x=vline_pos, color='black', linestyle='--', linewidth=1.5)
for cat, color in category_colors.items():
    plt.scatter([], [], color=color, label=cat)
plt.legend(title='Category', bbox_to_anchor=(1.02,1), loc='upper
left')
plt.xticks(np.arange(len(var_responses_ft_pp)),
            var_responses_ft_pp.index,
            rotation=90,
            fontsize=10)
plt.title("Variance responses_ft_pp")
plt.tight_layout()
plt.show()

```



Comment the results obtained for the variances (max 150 words):

One can observe that the range of the values after preprocessing is between 0 and 0.16, whereas before preprocessing it ranged between 0 and 2.5. This occurs because all features are now scaled within the interval [0, 1]. After preprocessing, the categorical features ("Smoking", "Alcohol", and "Internet usage") appear to be more relevant and contribute more to the PCA analysis. In contrast, the numerical features show similar behavior, as they have only been rescaled without altering their relative structure.

Write the code for computing all the n PCs of the two datasets, separately, and for visualizing the curves of cumulative explained variances:

```
# PCA
pca_responses_ft_enc = PCA()
pca_responses_ft_pp = PCA()

pca_responses_ft_enc.fit(responses_ft_enc)
pca_responses_ft_pp.fit(responses_ft_pp)

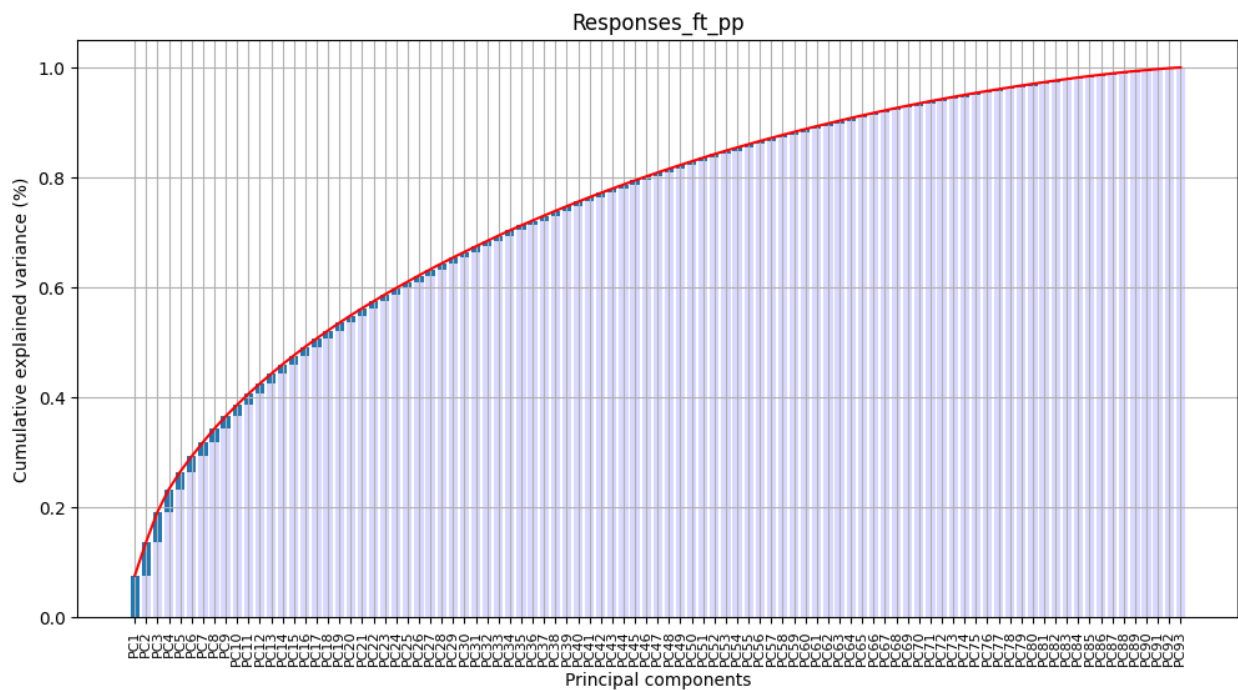
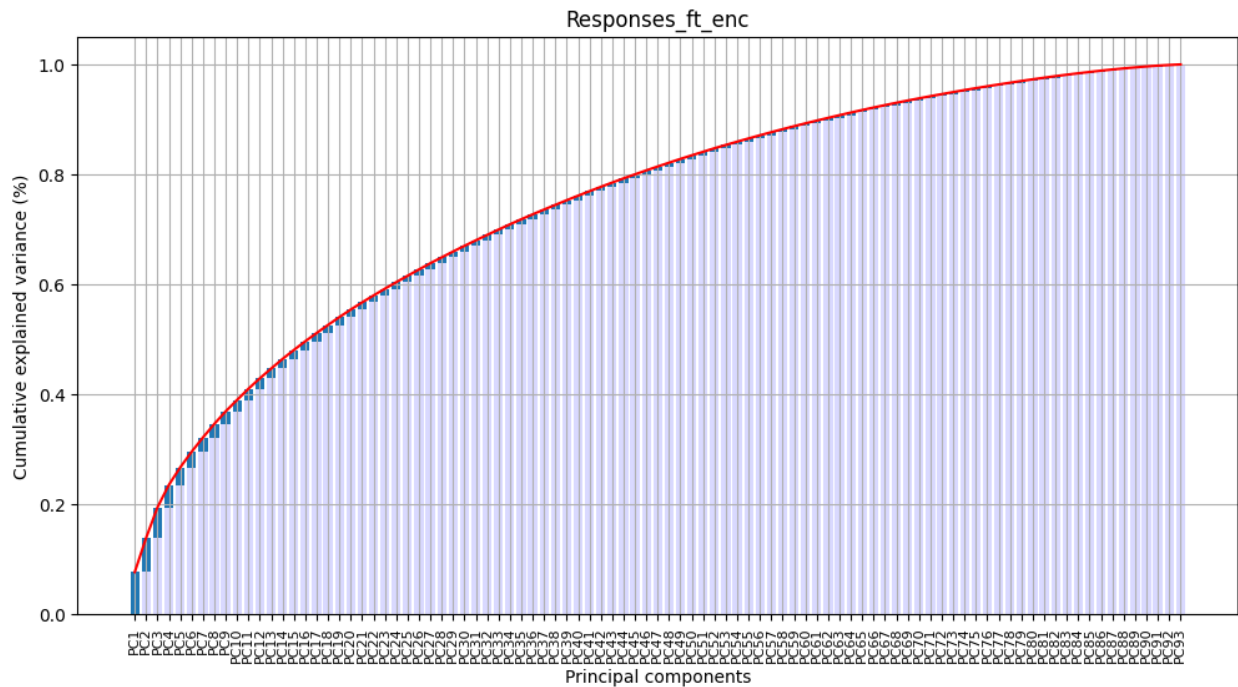
# plotting the cumulative explained variance - encoded dataset
plt.figure(figsize=(12,6))
plt.bar(np.arange(pca_responses_ft_enc.n_features_in_),
pca_responses_ft_enc.explained_variance_ratio_,
bottom=np.insert(np.cumsum(pca_responses_ft_enc.explained_variance_ratio_), 0, 0)[::-1])
```

```

plt.bar(np.arange(pca_responses_ft_enc.n_features_in_),
np.insert(np.cumsum(pca_responses_ft_enc.explained_variance_ratio_),
0, 0)[: -1], color='b', alpha=0.15)
plt.plot(np.cumsum(pca_responses_ft_enc.explained_variance_ratio_),
'r')
plt.title('Responses_ft_enc')
plt.xticks(ticks=np.arange(pca_responses_ft_enc.n_features_in_),
labels=[f'PC{i}' for i in range(1,
pca_responses_ft_enc.n_features_in_ + 1)],
rotation = 90,
fontsize=8)
plt.xlabel('Principal components')
plt.ylabel('Cumulative explained variance (%)')
plt.grid()
plt.show()

# plotting the cumulative explained variance - preprocessed dataset
plt.figure(figsize=(12,6))
plt.bar(np.arange(pca_responses_ft_pp.n_features_in_),
pca_responses_ft_pp.explained_variance_ratio_,
bottom=np.insert(np.cumsum(pca_responses_ft_pp.explained_variance_rati
o_), 0, 0)[: -1])
plt.bar(np.arange(pca_responses_ft_pp.n_features_in_),
np.insert(np.cumsum(pca_responses_ft_pp.explained_variance_ratio_), 0,
0)[: -1], color='b', alpha=0.15)
plt.plot(np.cumsum(pca_responses_ft_pp.explained_variance_ratio_),
'r')
# plt.hlines(y=.33, xmin=0, xmax=7, color='black', linestyle='--')
this is for the next exercise
plt.title('Responses_ft_pp')
plt.xticks(ticks=np.arange(pca_responses_ft_pp.n_features_in_),
labels=[f'PC{i}' for i in range(1,
pca_responses_ft_pp.n_features_in_ + 1)],
rotation = 90,
fontsize=8)
plt.xlabel('Principal components')
plt.ylabel('Cumulative explained variance (%)')
plt.grid()
plt.show()

```



Comment the results obtained for the cumulative explained variances, knowing the values in the datasets and the features' variances (max 150 words):

The two cumulative explained variance curves are very similar, indicating that the inclusion of the three categorical features does not significantly affect the overall variance captured by the principal components: their contribution remains limited when compared to the total number of features in the dataset (93) and their respective variances. In this case, the numerical features

account for the majority of the variance. This suggests that the added categorical features have a marginal impact on the global variance structure captured by PCA.

Exercise 3. Dimensionality Reduction and PC Interpretation

In this exercise, you have to do the following operations:

1. For the dataset *responses_ft_pp*, compute a new PCA for performing a dimensionality reduction with respect to m dimensions. The value of m must be
$$m = \min\{m', 5\},$$
 where m' is the value required for obtaining 33% of the total variance.
2. Visualize all the PCs as barplots and give an interpretation and a name to them, **motivating your choices**.
3. Visualize the score graph. If $m > 3$, plot the score graph with respect to the first 3 PCs. All the **plots must show the names of the PCs (given at the previous step) on the axes** for better understanding the results.

Write the code for computing the new PCA, for visualizing the m PCs as barplots:

```
# m = 5 due to the fact that m' = 8
m=5
pca_responses_ft_pp_m = PCA(n_components=m)
pca_responses_ft_pp_m.fit(responses_ft_pp)

round_expl_var_ratio =
np.round(pca_responses_ft_pp_m.explained_variance_ratio_.sum()*100,
decimals=2)

# eps is a threshold to observe the most relevant features for each PC
eps = np.sqrt(1/pca_responses_ft_pp_m.n_features_in_)

X = responses_ft_pp

# PCs' explained variance
plt.figure(figsize=(6,6))
plt.bar(range(1,m+1),pca_responses_ft_pp_m.explained_variance_ratio_)
plt.title(f"PCs' EXPLAINED VARIANCE ({round_expl_var_ratio}% OF
TOT.EXPL.VAR.)")
plt.xticks(np.arange(1,m+1),
           labels=[f'PC{i}' for i in range(1,m+1)],
           rotation=45)
plt.xlabel('Principal components')
plt.ylabel('Percentage of explained variance')
plt.grid(True, linestyle='--',linewidth=0.5,alpha=0.7)
plt.tight_layout()
plt.show()
```

```

# PCs barplots
colors = [column_color.get(col, 'gray') for col in X.columns]

for ii in range(m):
    plt.figure(figsize=(18, 6))
    plt.bar(np.arange(pca_responses_ft_pp_m.n_features_in_),
            pca_responses_ft_pp_m.components_[ii, :], color=colors)
    plt.plot([-0.5, pca_responses_ft_pp_m.n_features_in_ - 0.5],
             [eps, eps], 'red')
    plt.plot([-0.5, pca_responses_ft_pp_m.n_features_in_ - 0.5], [-
eps, -eps], 'red')
    plt.axvline(x=vline_pos, color='black', linestyle='--',
linewidth=1.5)
    for cat, color in category_colors.items():
        plt.scatter([], [], color=color, label=cat)
    plt.legend(title='Category', bbox_to_anchor=(1.02, 1), loc='upper
left')
    plt.xticks(ticks=np.arange(pca_responses_ft_pp_m.n_features_in_),
              labels=X.columns.to_list(),
              rotation=90)
    plt.title(f"PC{ii+1} loadings")
    plt.grid(alpha=0.3)
    plt.tight_layout()
    plt.show()

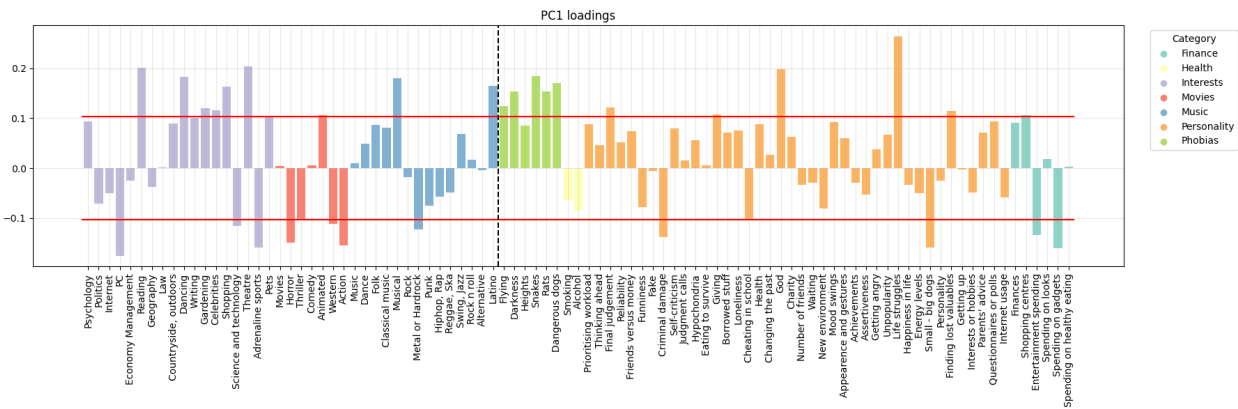
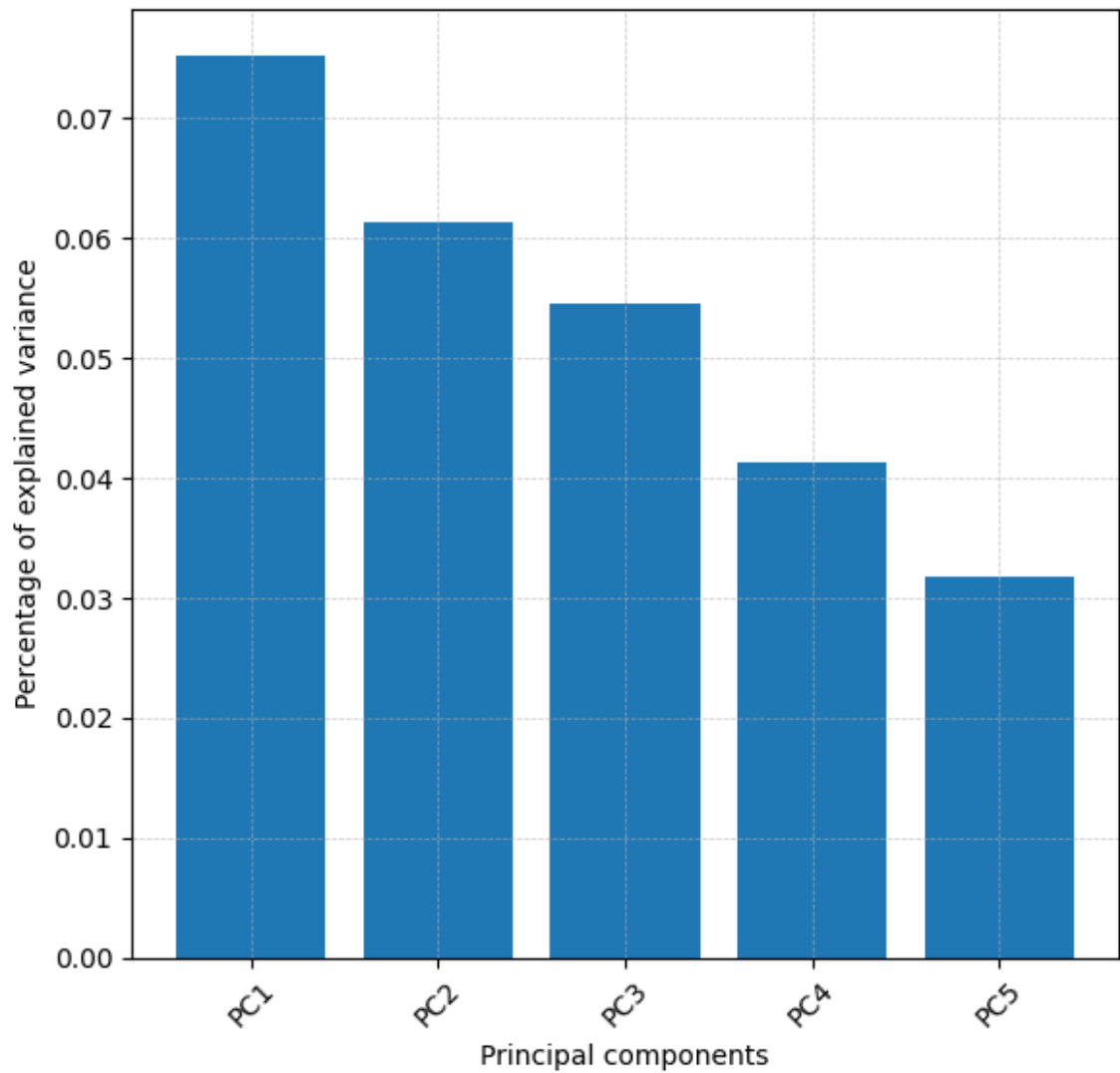
    ind_great_pos_PCii =
np.argmaxwhere(pca_responses_ft_pp_m.components_[ii, :] >= eps).flatten()
    ind_great_neg_PCii =
np.argmaxwhere(pca_responses_ft_pp_m.components_[ii, :] <= -
eps).flatten()

    great_pos_PCii = [X.columns[i] for i in ind_great_pos_PCii]
    great_neg_PCii = [X.columns[i] for i in ind_great_neg_PCii]

    print('')
    print(f'***** PC{ii+1} *****')
    print(f'HIGH-VALUED POSITIVE COMPONENTS: {great_pos_PCii}')
    print('')
    print(f'HIGH-VALUED NEGATIVE COMPONENTS: {great_neg_PCii}')
    print('*****')
    print('')

```

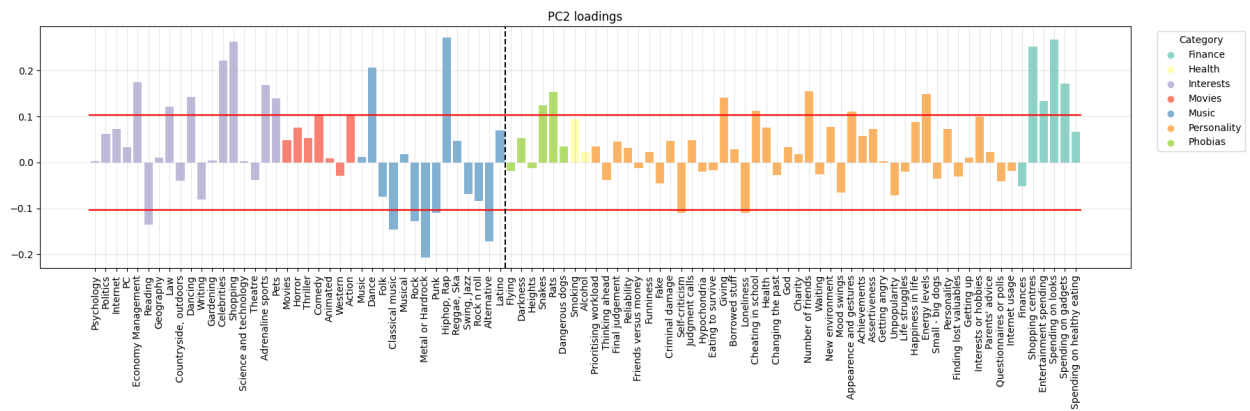
PCs' EXPLAINED VARIANCE (26.44% OF TOT.EXPL.VAR.)



***** PC1 *****

HIGH-VALUED POSITIVE COMPONENTS: ['Reading', 'Dancing', 'Gardening', 'Celebrities', 'Shopping', 'Theatre', 'Animated', 'Musical', 'Latino', 'Flying', 'Darkness', 'Snakes', 'Rats', 'Dangerous dogs', 'Final judgement', 'Giving', 'God', 'Life struggles', 'Finding lost valuables', 'Shopping centres']

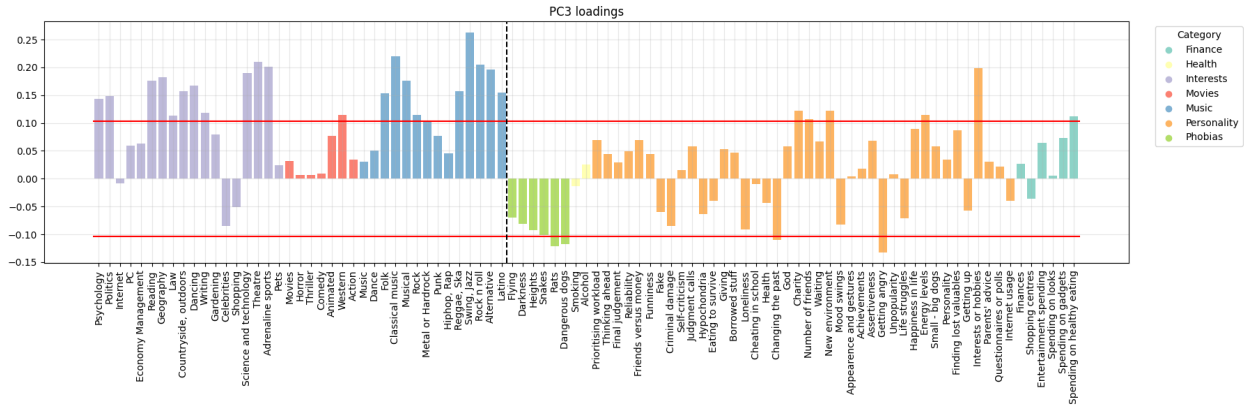
HIGH-VALUED NEGATIVE COMPONENTS: ['PC', 'Science and technology', 'Adrenaline sports', 'Horror', 'Western', 'Action', 'Metal or Hardrock', 'Criminal damage', 'Small - big dogs', 'Entertainment spending', 'Spending on gadgets']



***** PC2 *****

HIGH-VALUED POSITIVE COMPONENTS: ['Economy Management', 'Law', 'Dancing', 'Celebrities', 'Shopping', 'Adrenaline sports', 'Pets', 'Dance', 'Hiphop, Rap', 'Snakes', 'Rats', 'Giving', 'Cheating in school', 'Number of friends', 'Appearance and gestures', 'Energy levels', 'Shopping centres', 'Entertainment spending', 'Spending on looks', 'Spending on gadgets']

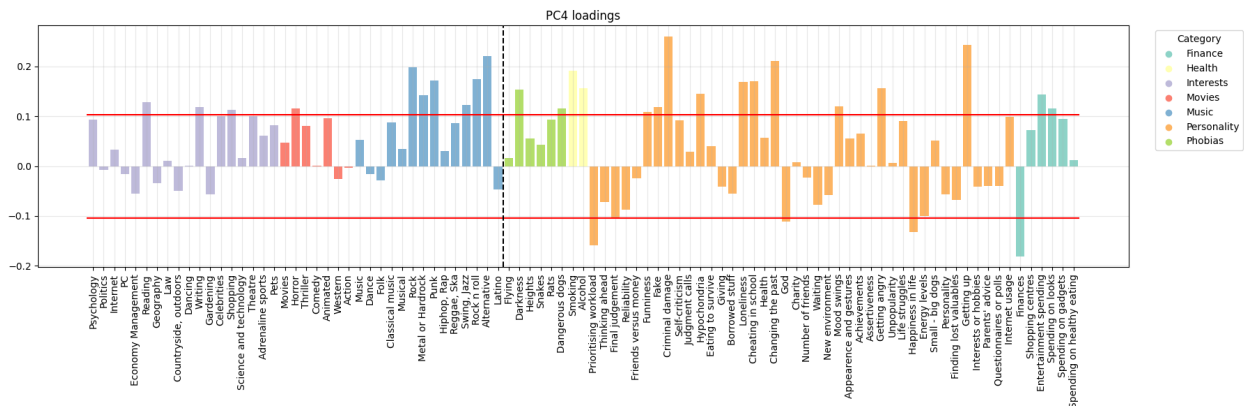
HIGH-VALUED NEGATIVE COMPONENTS: ['Reading', 'Classical music', 'Rock', 'Metal or Hardrock', 'Punk', 'Alternative', 'Self-criticism', 'Loneliness']



***** PC3 *****

HIGH-VALUED POSITIVE COMPONENTS: ['Psychology', 'Politics', 'Reading', 'Geography', 'Law', 'Countryside, outdoors', 'Dancing', 'Writing', 'Science and technology', 'Theatre', 'Adrenaline sports', 'Western', 'Folk', 'Classical music', 'Musical', 'Rock', 'Metal or Hardrock', 'Reggae, Ska', 'Swing, Jazz', 'Rock n roll', 'Alternative', 'Latino', 'Charity', 'Number of friends', 'New environment', 'Energy levels', 'Interests or hobbies', 'Spending on healthy eating']

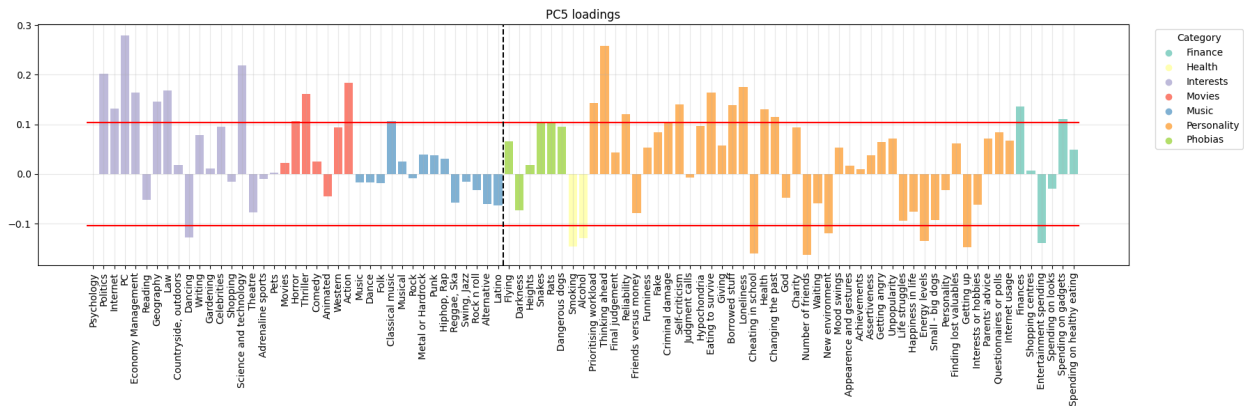
HIGH-VALUED NEGATIVE COMPONENTS: ['Rats', 'Dangerous dogs', 'Changing the past', 'Getting angry']



***** PC4 *****

HIGH-VALUED POSITIVE COMPONENTS: ['Reading', 'Writing', 'Shopping', 'Horror', 'Rock', 'Metal or Hardrock', 'Punk', 'Swing, Jazz', 'Rock n roll', 'Alternative', 'Darkness', 'Dangerous dogs', 'Smoking', 'Alcohol', 'Funniness', 'Fake', 'Criminal damage', 'Hypochondria', 'Loneliness', 'Cheating in school', 'Changing the past', 'Mood swings', 'Getting angry', 'Getting up', 'Entertainment spending', 'Spending on looks']

HIGH-VALUED NEGATIVE COMPONENTS: ['Prioritising workload', 'God', 'Happiness in life', 'Finances']



***** PC5 *****

HIGH-VALUED POSITIVE COMPONENTS: ['Politics', 'Internet', 'PC', 'Economy Management', 'Geography', 'Law', 'Science and technology', 'Horror', 'Thriller', 'Action', 'Classical music', 'Snakes', 'Rats', 'Prioritising workload', 'Thinking ahead', 'Reliability', 'Self-criticism', 'Eating to survive', 'Borrowed stuff', 'Loneliness', 'Health', 'Changing the past', 'Finances', 'Spending on gadgets']

HIGH-VALUED NEGATIVE COMPONENTS: ['Dancing', 'Smoking', 'Alcohol', 'Cheating in school', 'Number of friends', 'New environment', 'Energy levels', 'Getting up', 'Entertainment spending']

For each PC, write the name you assigned to it and a brief interpretation that motivate the choice (max 100 words per PC):

PC1: (-) Tech-Action Orientation vs (+) Emotional-Spiritual

On the negative pole, there are technological interests, action-oriented preferences, financial focus, and low emotionality traits in personality. This pole is more strongly connected to technology and action. On the positive pole, there are cultural and relaxing interests, musical preferences, 'Life struggles' in personality category, and phobias. This dimension is therefore more connected with humanity, spirituality and emotional sensitivity.

PC2: (-) Introverted Reflection vs (+) Social-Consumer Expressiveness

On the negative pole, there is a prevalence of music preferences, 'Reading' in personality category, and some introspective personality traits, which suggest an introverted profile. On the positive pole, there are social and consumer-oriented interests such as 'Celebrities', 'Shopping',

'Hiphop, Rap' and 'Shopping centres', 'Spending on looks'. This direction captures consumer tendencies and social expressiveness.

PC3: (-) Emotional Reactivity vs (+) Entertainment

On the negative pole, there are phobias, 'Changing the past' and 'Getting angry' in personality category, all of which are related to emotional reactivity. On the positive pole, there are almost all features connected with entertainment responses, indicating a strong orientation toward cultural and leisure activities.

PC4: (-) Stability vs (+) Transgression

On the negative pole, there are 'Finances' and indicators of life balance and well-being in the personality category. This pole can therefore be summarized as stability. On the positive pole, there are music preferences, health behaviors, and several rebellious personality traits, which characterize a dimension of transgression.

PC5: (-) Social Vitality vs (+) Rationality

On the negative pole, there is a combination of relational personality traits, health behaviors and dancing interests, which can be grouped under social vitality. On the positive pole, there are interests, movie preferences, and personality traits related to analytical thinking, self-control, and planning, which can be summarized as rationality.

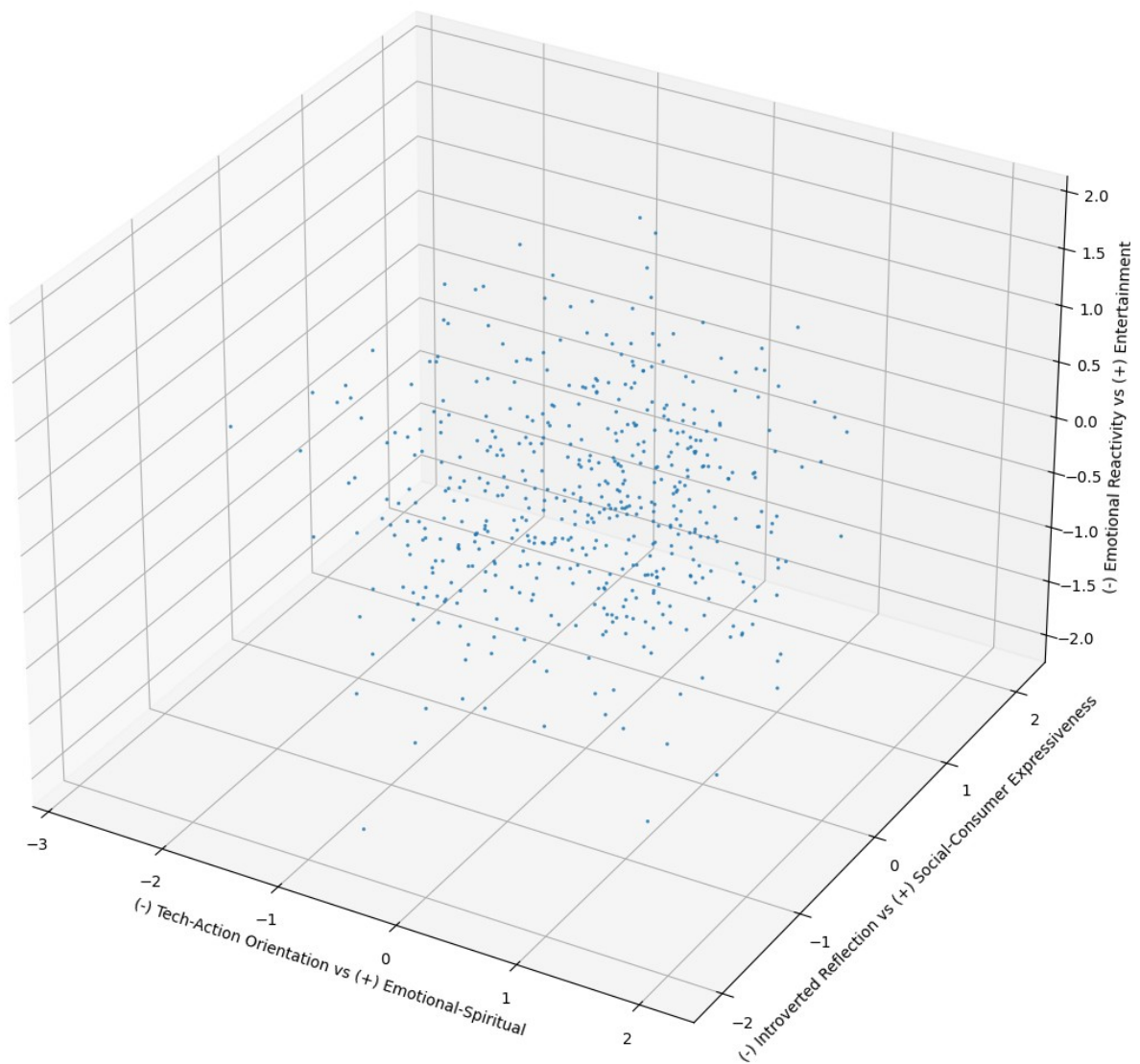
Write the code for visualizing the score graph (with PC names on the axis):

```
pc_names = ['(-) Tech-Action Orientation vs (+) Emotional-Spiritual',
            '(-) Introverted Reflection vs (+) Social-Consumer Expressiveness',
            '(-) Emotional Reactivity vs (+) Entertainment',
            '(-) Stability vs (+) Transgression',
            '(-) Social Vitality vs (+) Rationality']

# change of basis
Y = pca_responses_ft_pp_m.transform(X)
responses_ft_pca = pd.DataFrame(Y, columns=pc_names)

# 3D score graph
fig_res = plt.figure(figsize= (11,11))
ax = fig_res.add_subplot(111, projection='3d')
ax.scatter(Y[:, 0], Y[:, 1], Y[:, 2], s=2, alpha=0.7)
plt.title('RESPONSES - SCORE GRAPH 3D')
ax.set_xlabel(pc_names[0], fontsize=10)
ax.set_ylabel(pc_names[1], fontsize=10)
ax.set_zlabel(pc_names[2], fontsize=10)
plt.grid()
plt.tight_layout()
plt.show()
```

RESPONSES - SCORE GRAPH 3D



Exercise 4. k -Means

In this exercise, you have to do the following operations:

1. Transform the *responses_ft_pp* data into their m -dimensional representation via PCA. Store the transformed data in the variable *responses_ft_pca*;
2. Run the k -Means for clustering the data of *responses_ft_pca*. In particular, **use the silhouette score for identifying the best value for $k \in \{3, \dots, 10\}$** and show it by plotting how the score changes w.r.t. k .
3. Plot the score graph again, but add the centroids of the cluster and color the points according to their cluster.

4. Visualize the centroids coordinates as barplots and **give a name and an interpretation to them by exploiting the PC names.**

Write the code for performing the items of the list above:

```
# for identifying the best value for k
K = range(3,10)
seed = 42
sil_score= []

for k in K:
    km_responses = KMeans(n_clusters=k, random_state=seed)
    km_responses.fit(responses_ft_pca)
    silcoeff = silhouette_score(responses_ft_pca,
km_responses.labels_)
    sil_score.append(silcoeff)

print(f'Silhouette scores: {sil_score}\nThe best k is:
{sil_score.index(max(sil_score))+3}')

k = sil_score.index(max(sil_score))+3
km_responses = KMeans(n_clusters=k, random_state=seed)
km_responses.fit(responses_ft_pca)
silcoeff = silhouette_score(responses_ft_pca, km_responses.labels_)

# score graph - 3D
sg_km = plt.figure(figsize= (10,10))
ax = sg_km.add_subplot(111, projection='3d')
ax.scatter(Y[:, 0], Y[:, 1], Y[:, 2], s=2, c=km_responses.labels_,
alpha=0.7)
ax.scatter(km_responses.cluster_centers[:,0],km_responses.cluster_cen
ters[:,1],km_responses.cluster_centers[:,2], c='black', marker='o')
for kk in range(k):
    ax.text(km_responses.cluster_centers_[kk,0],
km_responses.cluster_centers_[kk,1],km_responses.cluster_centers_[kk,2
], f'cluster{kk+1}')
plt.title('RESPONSES - SCORE GRAPH')
ax.set_xlabel(pc_names[0])
ax.set_ylabel(pc_names[1])
ax.set_zlabel(pc_names[2])
plt.grid()
plt.tight_layout()
plt.show()

maxs_res_m = responses_ft_pca.max(axis=0)
mins_res_m = responses_ft_pca.min(axis=0)

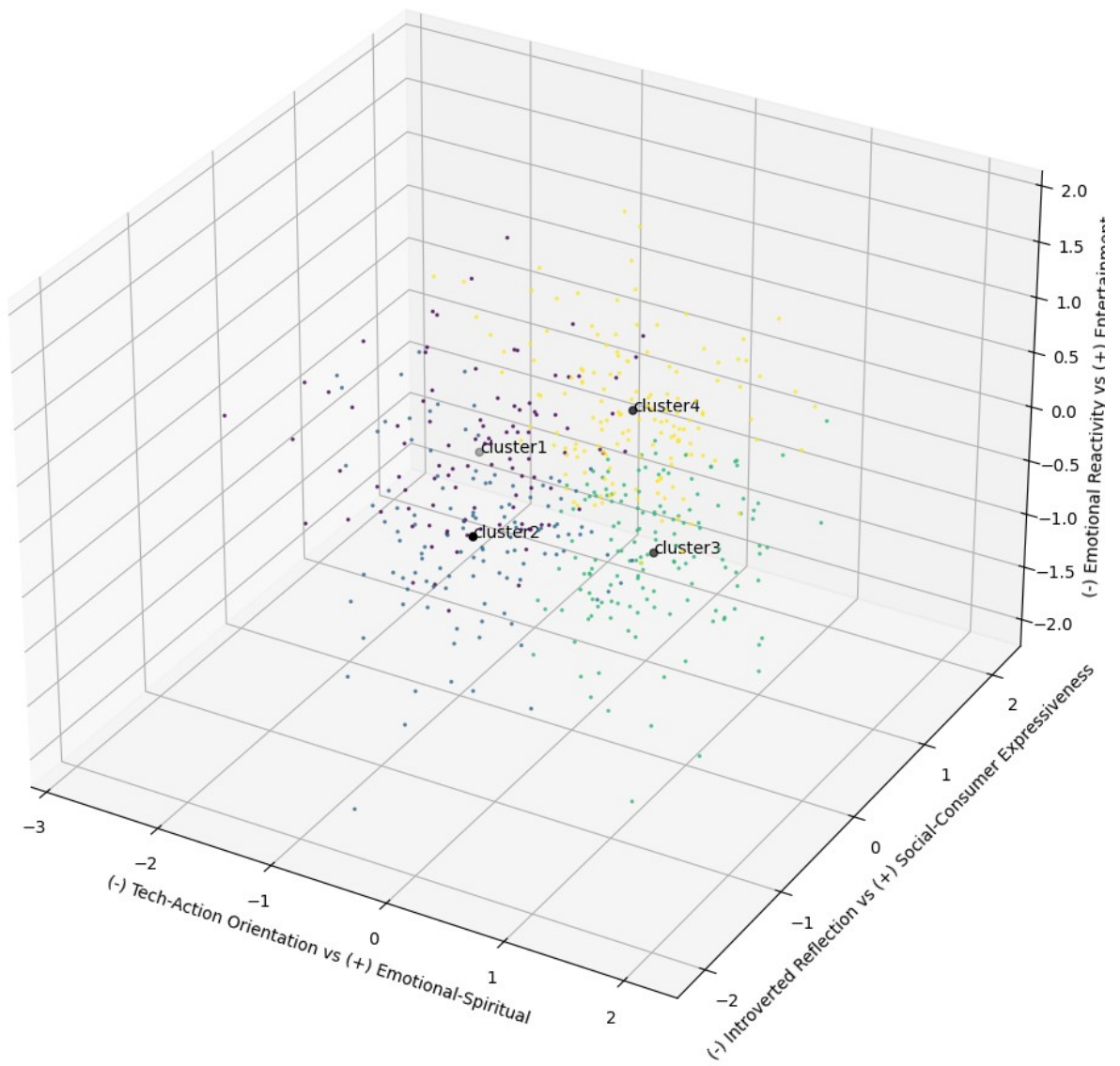
# reduced pc names
pc_names_red = [ '(-) Tech-Action vs (+) Emotional-Spiritual',
                  '(-) Introverted vs (+) Social-Consumer',
                  '(-) Emotional Reactivity vs (+) Entertainment',
```

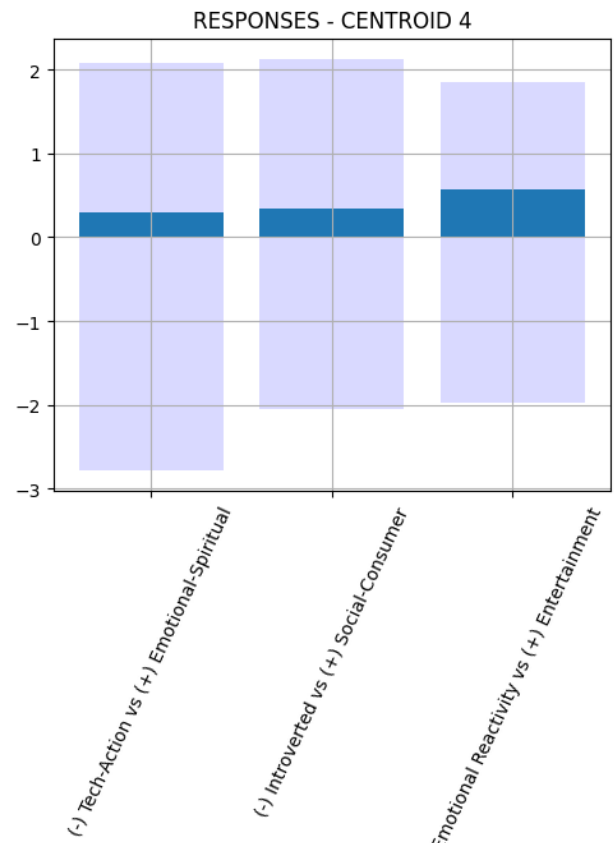
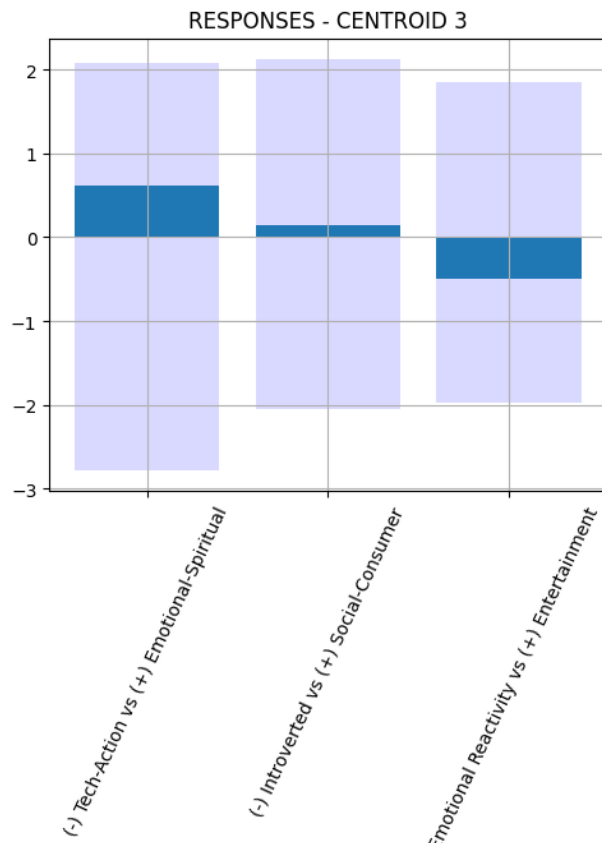
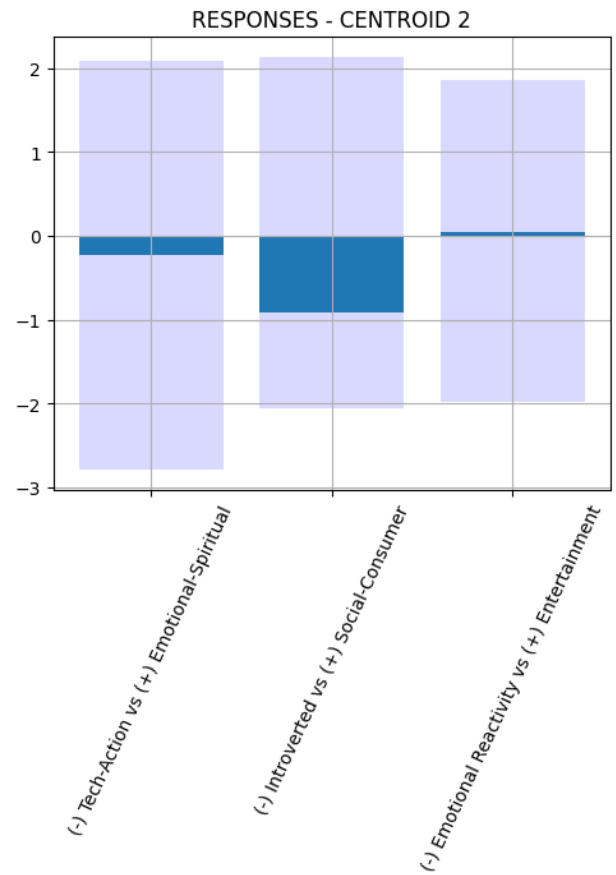
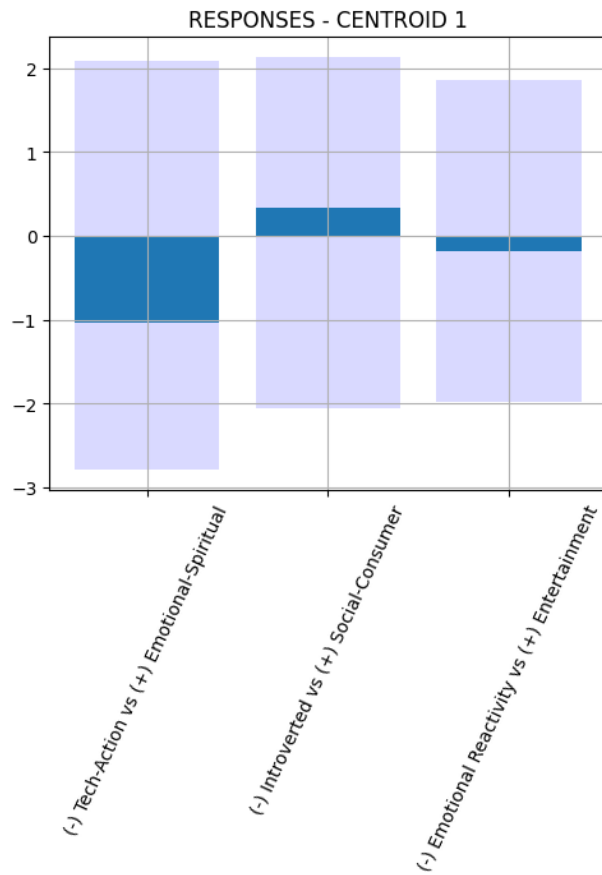
```
'(-) Stability vs (+) Transgression',  
'(-) Social Vitality vs (+) Rationality']
```

```
# centroids coordinates as barplots  
fig_centroids, ax_centroids = plt.subplots(2, 2, figsize=(10, 14))  
for ii in range(k):  
    ir= ii//2  
    ic= ii%2  
    n_pc = 3  
    ax_centroids[ir,ic].bar(np.arange(n_pc), maxs_res_m[:n_pc],  
color='blue', alpha=0.15)  
    ax_centroids[ir,ic].bar(np.arange(n_pc), mins_res_m[:n_pc],  
color='blue', alpha=0.15)  
    ax_centroids[ir,ic].bar(np.arange(n_pc),  
km_responses.cluster_centers_[ii, :n_pc])  
    ax_centroids[ir,ic].set_xticks(ticks=np.arange(n_pc))  
    ax_centroids[ir,ic].set_xticklabels(labels=pc_names_red[:n_pc],  
rotation=65)  
    ax_centroids[ir,ic].grid(visible=True, which='both')  
    ax_centroids[ir,ic].set_title(f'RESPONSES - CENTROID {ii+1}')  
plt.tight_layout()  
plt.show()
```

```
Silhouette scores: [0.15941958881122792, 0.16567595792947243,  
0.1541770712552679, 0.15952009869183723, 0.15446343791747716,  
0.15287589037969804, 0.1561359015904033]  
The best k is: 4
```

RESPONSES - SCORE GRAPH





For each Centroid, write the name you assigned to it and a brief interpretation that motivate the choice by exploiting the PC names(max 100 words per centroid):

Centroid 1: Tech-Action Orientation

This centroid is mainly associated with the negative pole of PC1, which represents a tech-action orientation. Individuals in this cluster show a stronger focus on technology, action-oriented activities, and pragmatic interests, combined with lower emotional and spiritual sensitivity. This profile reflects a preference for instrumental, performance-driven behaviors rather than cultural or emotionally expressive ones.

Centroid 2: Introverted Reflection

This centroid is primarily defined by the negative pole of PC2, corresponding to introverted reflection. It captures individuals characterized by introspective traits, solitary cultural interests such as reading and non-popular music, and higher self-reflection. Social and consumer-oriented behaviors are less prominent in this cluster.

Centroid 3: Emotional Sensitivity and Reactivity

This centroid is characterized by the positive pole of PC1 (Emotional–Spiritual) and the negative pole of PC3 (Emotional Reactivity). It represents individuals with high emotional sensitivity, spiritual, and greater vulnerability to emotional responses. This cluster reflects a strong emotional engagement with personal experiences.

Centroid 4: Social and Cultural Entertainment

This centroid is mainly associated with the positive pole of PC3, which captures entertainment and cultural aspects. Individuals in this cluster show a strong orientation toward leisure activities, cultural consumption, music, movies, and diverse interests. There is also a contribution from the positive pole of PC1 and PC2, which are related to an emotional sensitivity and social-consumer tendencies.

Exercise 5. Cluster External Evaluations

In this exercise, you have to do the following operations:

1. Select a subset meaningful labels for performing an external evaluation of the clustering results.
2. For each selected label, visualize the distribution of the label in each cluster and in the whole dataset. Moreover, visualize the clusters in separated score-graphs, coloring the points according to the label values.

List the Labels you consider meaningful for an external cluster evaluation and motivate your choice (max 50 words per label):

The labels that appear meaningful for external cluster evaluation are 'Gender' and 'Weight', as their distributions vary across clusters. Other labels show similar proportions in all clusters and therefore do not provide informative differences for assessing cluster quality.

Write the code for the visualizations cited in item 2 above:

```
centroids_names= ['Tech-Action Orientation','Introverted
Reflection','Emotional Sensitivity and Reactivity','Social and
Cultural Entertainment']
selected_labels = ['Gender','Weight']

for lb in selected_labels:
    lb_values, lb_counts = np.unique(responses_lb[lb].values,
return_counts=True)
    if lb == 'Weight':
        n_colors_hist = responses_lb[lb].max() -
responses_lb[lb].min() +1
        cmap_hist = plt.get_cmap('viridis')
        color_hist = cmap_hist(np.linspace(0, 1, n_colors_hist))
        color_scatter = [color_hist[ll- responses_lb[lb].min(), :3] for
ll in responses_lb[lb].values]
    else:
        n_colors_hist = lb_values.size
        cmap_hist = plt.get_cmap('tab10')
        color_scatter_dict = {lb_values[ii]: ii for ii in
range(len(lb_values))}
        color_scatter = [cmap_hist(color_scatter_dict[ll]) for ll in
responses_lb[lb].values]
        color_hist = cmap_hist(np.arange(0, n_colors_hist))

    color_hist = color_hist[:, :3]

    # global plots
    fig_lb = plt.figure(figsize= (12,6))

    ax1 = fig_lb.add_subplot(1,2,1, projection='3d')
    scatter1 = ax1.scatter(Y[:, 0], Y[:, 1], Y[:, 2], s=2, c=color_scatter,
alpha=0.5)
    ax1.set_title(f'SCORE GRAPH W.R.T. LABEL {lb}')
    ax1.set_xlabel(pc_names_red[0], fontsize=8)
    ax1.set_ylabel(pc_names_red[1], fontsize=8)
    ax1.set_zlabel(pc_names_red[2], fontsize=8)
    ax1.grid(True, linestyle='--', linewidth=0.5, alpha=0.5)
    plt.tight_layout()

    ax2 = fig_lb.add_subplot(1,2,2)
    ax2.bar(lb_values, lb_counts, color = color_hist)
    if lb == 'Weight' or lb == 'Age':
        mean_value = np.average(lb_values, weights=lb_counts)
        ax2.axvline(x=mean_value, linestyle='--', linewidth=2,
label=f'Mean = {mean_value:.2f}')
        ax2.legend()
    else:
        for x, count in zip(lb_values, lb_counts):
```

```

ax2.text(x, count, str(count), ha='center', va='bottom', fontsize=9)
ax2.set_title(f'HISTOGRAM OF LABEL {lb}')
ax2.set_xlabel('Categories')
ax.set_ylabel('Values')
ax2.grid(True, linestyle='--', linewidth=0.5, alpha=0.5)
plt.tight_layout()

plt.show()

# cluster plots
for cc in range(km_responses.cluster_centers_.shape[0]):
    where_cc = np.argwhere(km_responses.labels_ == cc).flatten()
    cc_values_, cc_counts_ =
np.unique(responses_lb[lb].values[where_cc], return_counts=True)
    new_counts = np.zeros_like(lb_values, dtype=int)

    for val, count in zip(cc_values_, cc_counts_):
        idx = np.where(lb_values == val)[0]
        new_counts[idx] = count

    cc_values_ = lb_values
    cc_counts_ = new_counts

    fig_cc = plt.figure(figsize= (12,6))

    ax1 = fig_cc.add_subplot(1,2,1, projection='3d')
    scat1 = ax1.scatter(Y[where_cc, 0], Y[where_cc, 1],
Y[where_cc, 2], s=2, c=[color_scat[wcc] for wcc in where_cc],
alpha=0.5)
    ax1.set_title(f'SCORE GRAPH W.R.T. LABEL {lb}\n CLUSTER {cc+1}
- {centroids_names[cc]}')
    ax1.set_xlabel(pc_names_red[0], fontsize=8)
    ax1.set_ylabel(pc_names_red[1], fontsize=8)
    ax1.set_zlabel(pc_names_red[2], fontsize=8)
    ax1.grid(True, linestyle='--', linewidth=0.5, alpha=0.5)
    plt.tight_layout()

    ax2 = fig_cc.add_subplot(1,2,2)
    ax2.bar(lb_values, cc_counts_, color = color_hist)
    if lb == 'Weight' or lb == 'Age':
        mean_cc = np.average(lb_values, weights=cc_counts_)
        ax2.axvline(x=mean_cc, linestyle='--', linewidth=2,
label=f'Mean = {mean_cc:.2f}')
        ax2.legend()
    else:
        vals, counts = np.unique(responses_lb[lb].values,
return_counts=True)
        global_perc = counts / counts.sum()
        for x, count in zip(lb_values, cc_counts_):

```

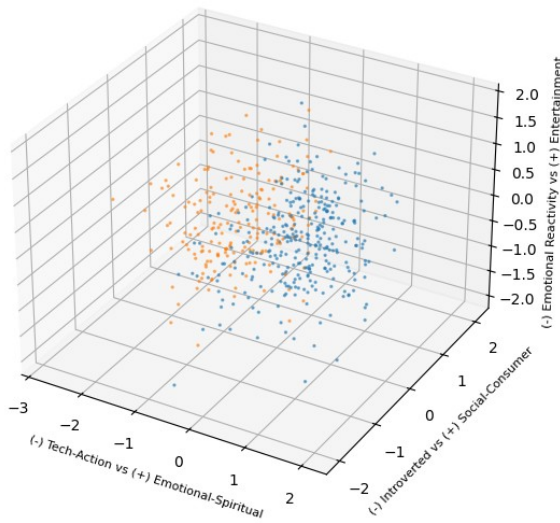
```

ax2.text(x,count,str(count),ha='center',va='bottom',fontsize=9)
ax2.set_title(f'HISTOGRAM OF LABEL {lb}\n CLUSTER {cc+1} - {centroids_names[cc]}')
ax2.set_xlabel('Categories')
ax2.set_ylabel('Values')
ax2.grid(True, linestyle='--', linewidth=0.5, alpha=0.5)
plt.tight_layout()

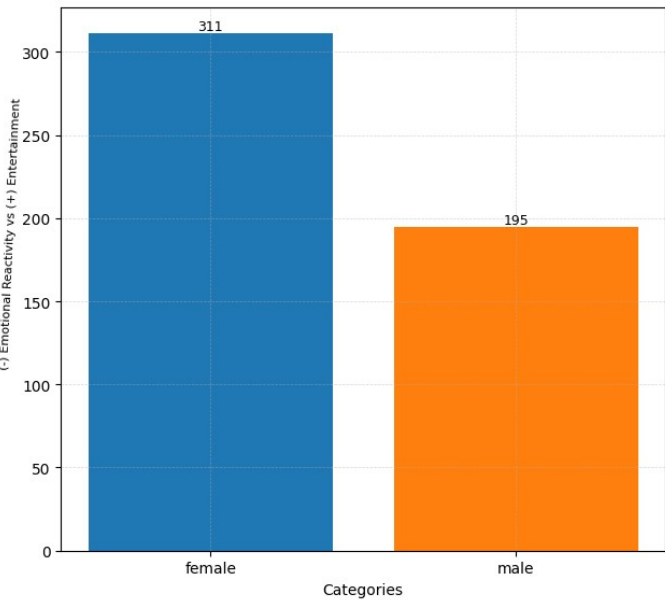
plt.show()

```

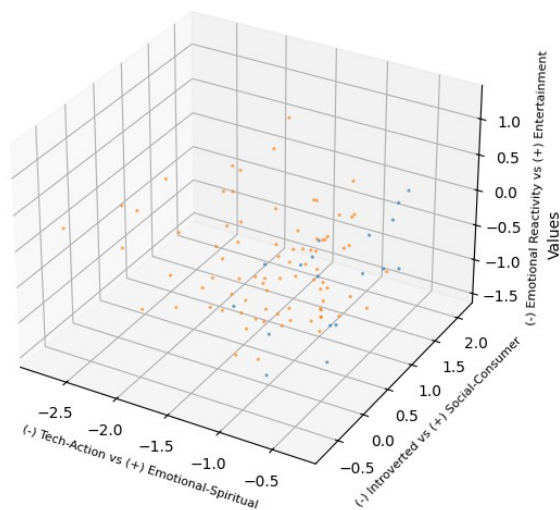
SCORE GRAPH W.R.T. LABEL Gender



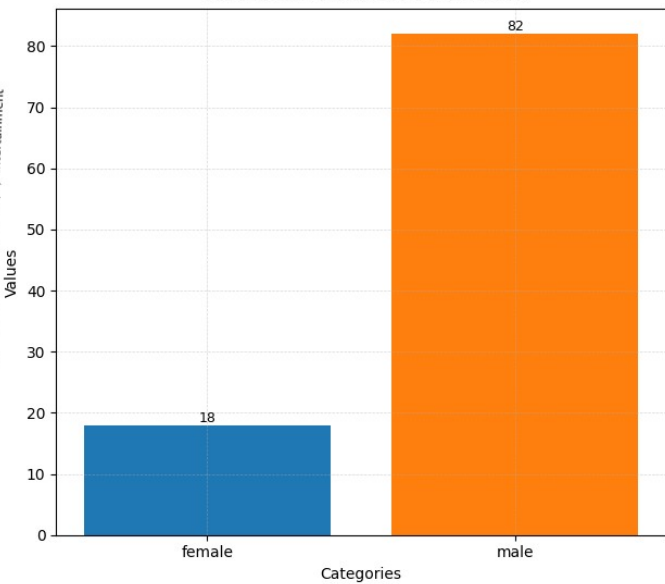
HISTOGRAM OF LABEL Gender



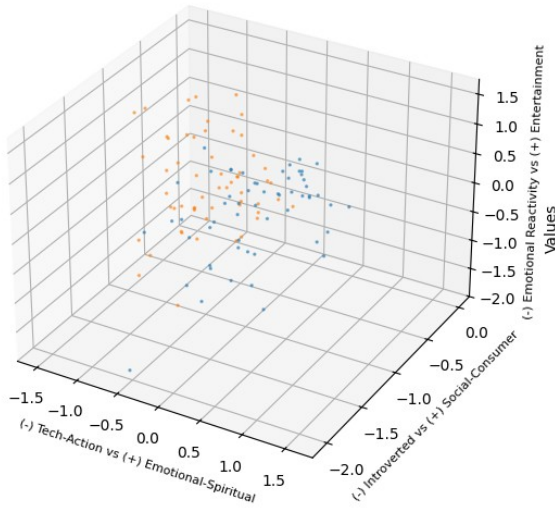
SCORE GRAPH W.R.T. LABEL Gender
CLUSTER 1 - Tech-Action Orientation



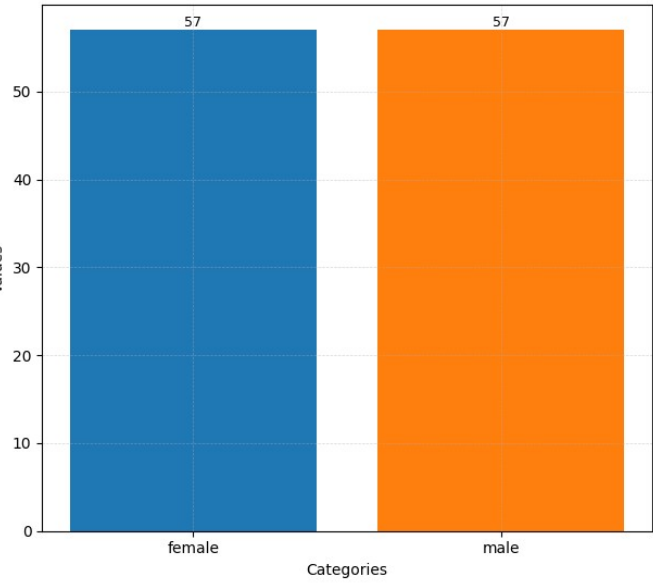
HISTOGRAM OF LABEL Gender
CLUSTER 1 - Tech-Action Orientation



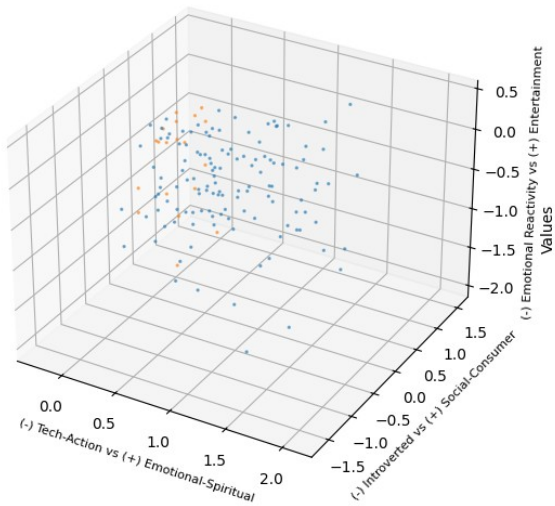
SCORE GRAPH W.R.T. LABEL Gender
CLUSTER 2 - Introverted Reflection



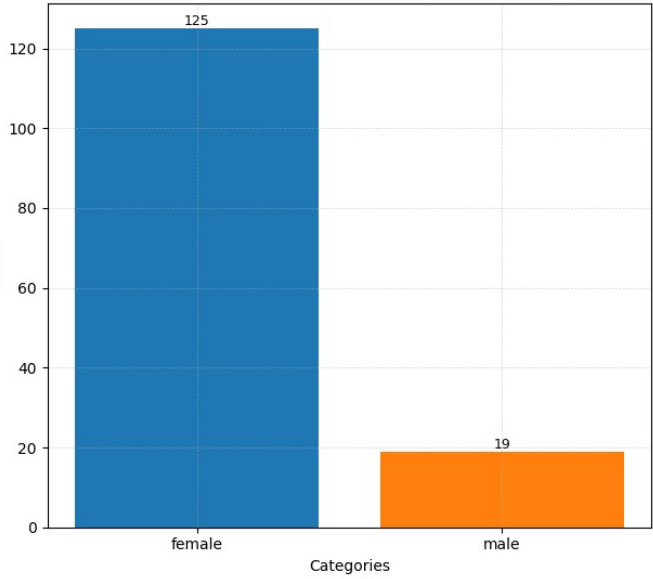
HISTOGRAM OF LABEL Gender
CLUSTER 2 - Introverted Reflection



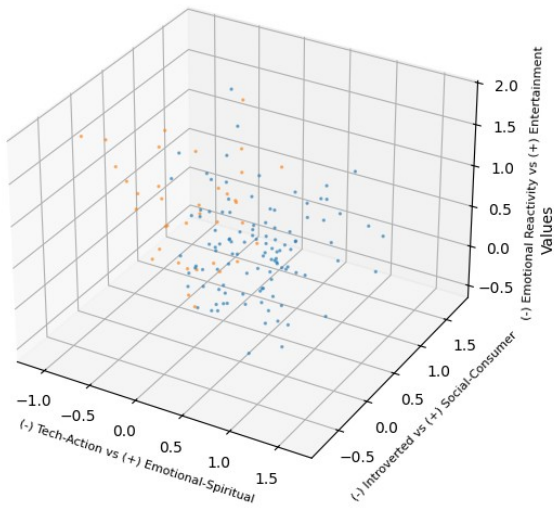
SCORE GRAPH W.R.T. LABEL Gender
CLUSTER 3 - Emotional Sensitivity and Reactivity



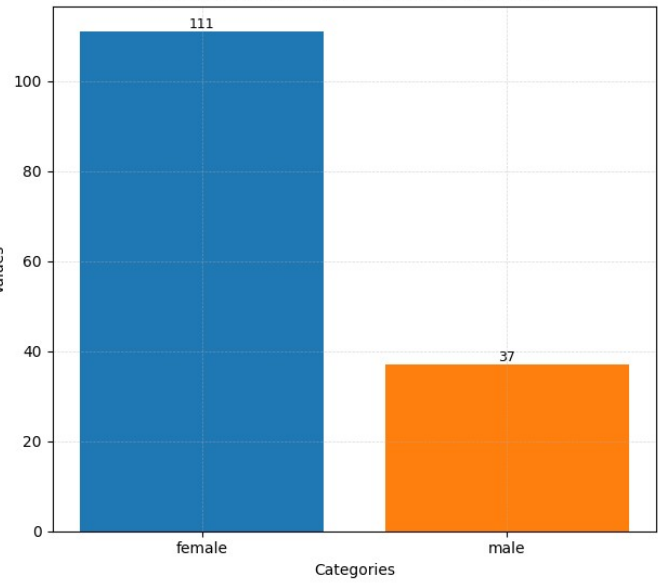
HISTOGRAM OF LABEL Gender
CLUSTER 3 - Emotional Sensitivity and Reactivity



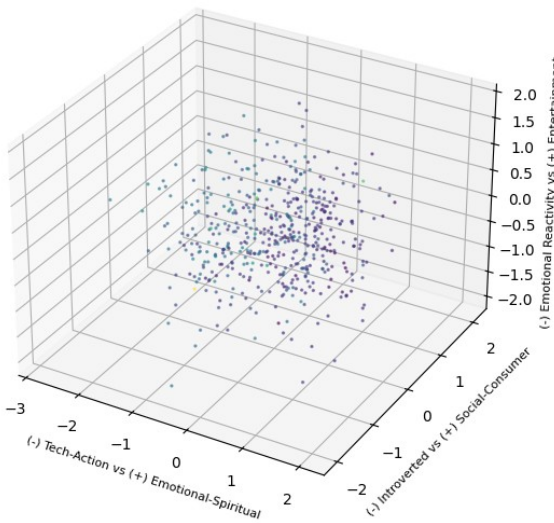
SCORE GRAPH W.R.T. LABEL Gender
CLUSTER 4 - Social and Cultural Entertainment



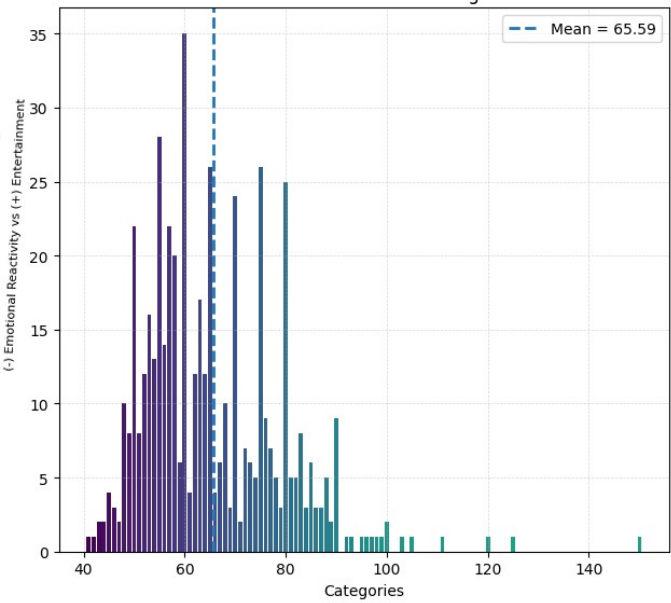
HISTOGRAM OF LABEL Gender
CLUSTER 4 - Social and Cultural Entertainment



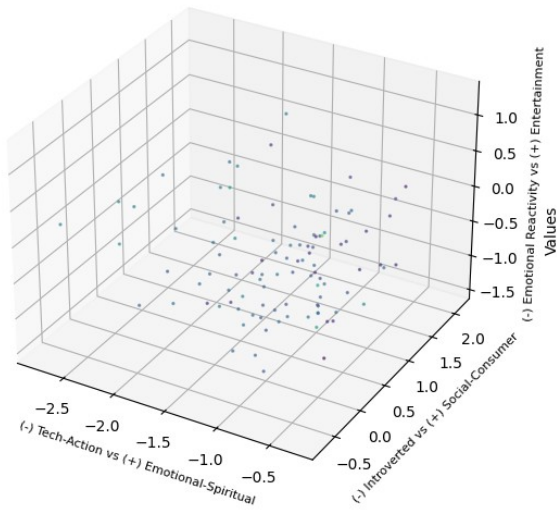
SCORE GRAPH W.R.T. LABEL Weight



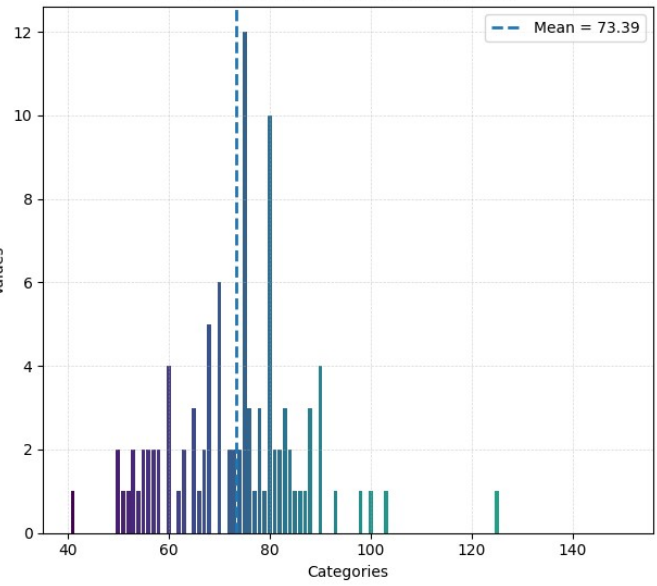
HISTOGRAM OF LABEL Weight



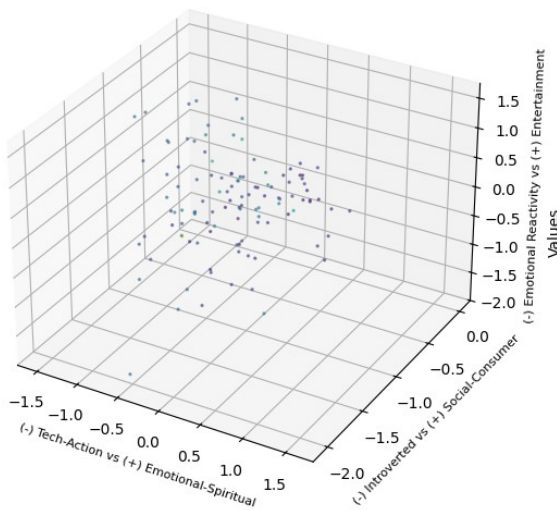
SCORE GRAPH W.R.T. LABEL Weight
CLUSTER 1 - Tech-Action Orientation



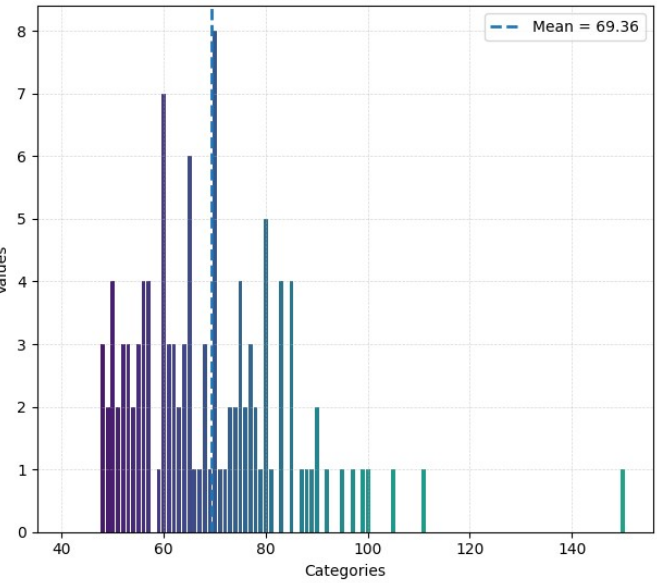
HISTOGRAM OF LABEL Weight
CLUSTER 1 - Tech-Action Orientation



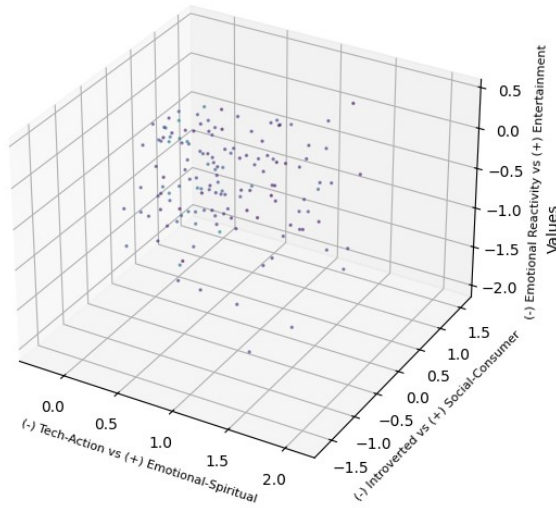
SCORE GRAPH W.R.T. LABEL Weight
CLUSTER 2 - Introverted Reflection



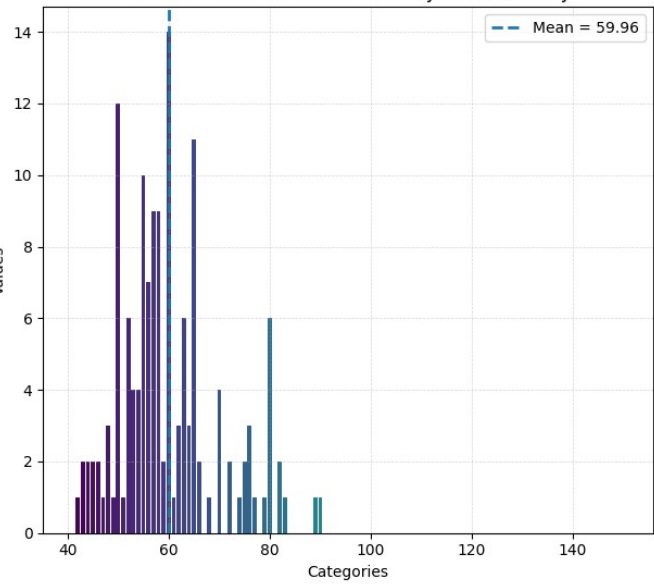
HISTOGRAM OF LABEL Weight
CLUSTER 2 - Introverted Reflection



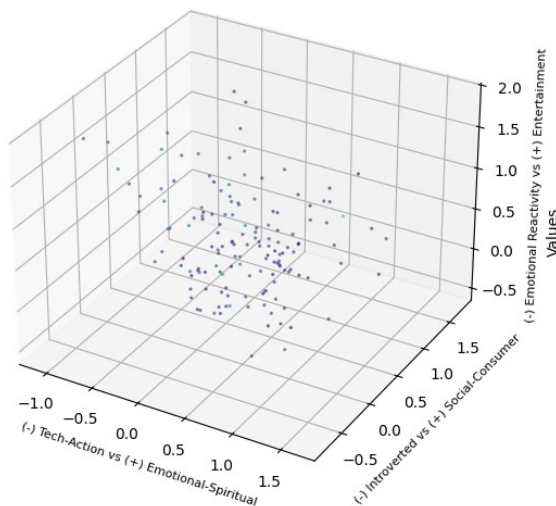
SCORE GRAPH W.R.T. LABEL Weight
CLUSTER 3 - Emotional Sensitivity and Reactivity



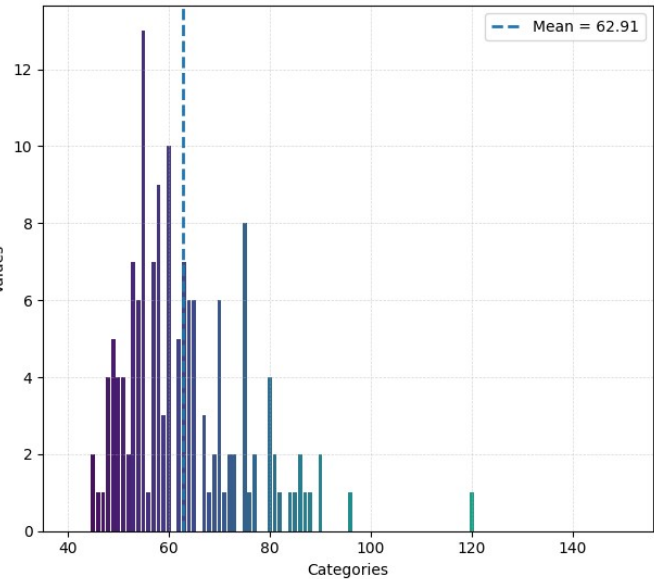
HISTOGRAM OF LABEL Weight
CLUSTER 3 - Emotional Sensitivity and Reactivity



SCORE GRAPH W.R.T. LABEL Weight
CLUSTER 4 - Social and Cultural Entertainment



HISTOGRAM OF LABEL Weight
CLUSTER 4 - Social and Cultural Entertainment



For each selected label, comment the results observed in the visualizations (max 100 words per label):

Gender:

In general, practical characteristics tend to be associated more with males, while emotional traits are more associated with females. This pattern is also observable in these clusters: the 'Tech-Action Orientation' cluster has a higher percentage of males, whereas the 'Emotional Sensitivity and Reactivity' cluster has a higher percentage of females compared to the global distribution. The 'Introverted Reflection' cluster has equal number of males and females, but

men are overrepresented relative to the overall population. Finally, the 'Social and Cultural Entertainment' cluster is largely neutral regarding gender.

Weight:

Considering the analysis by gender, there appears to be a correlation between weight and gender. Clusters with a higher proportion of males tend to have an average weight above the global mean, while clusters with a predominance of females tend to have an average weight below the global mean. This pattern reflects the overall differences in weight between genders, and it is observable in the cluster distributions, showing that the gender composition within each cluster influences the average weight values relative to the total population.

Exercise 6. Cluster Internal Evaluations

In this exercise, you have to do the following operations:

1. For each cluster, measure the corresponding average silhouette score
2. Visualize the silhouette of the clusters and the general one of the clustering and compare them

Write the code for computing the silhouette scores and for visualizing them:

```
silscores = silhouette_samples(Y, km_responses.labels_)
cluster_silscores = [np.mean(silscores[km_responses.labels_ == kk])
for kk in range(k)]

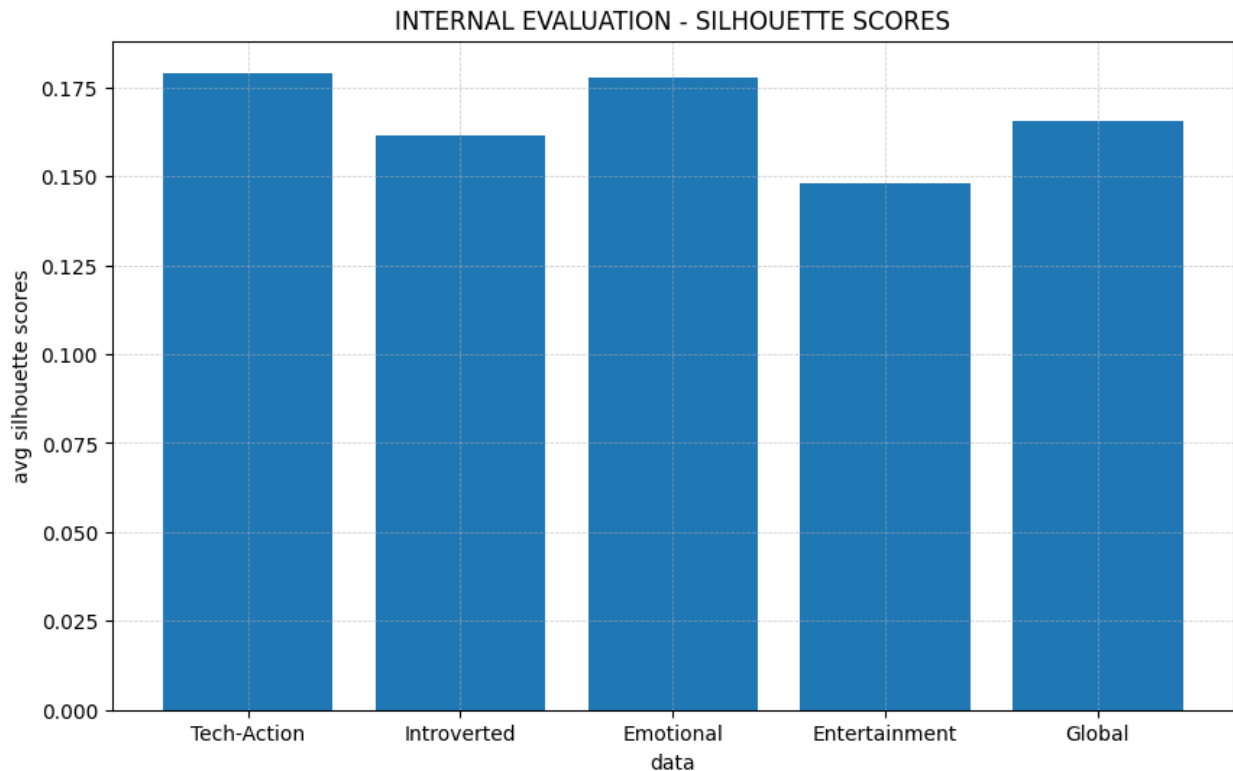
centroids_names_red= ['Tech-
Action', 'Introverted', 'Emotional', 'Entertainment']

# avg silhouette scores
silcoeffs_df = pd.DataFrame(np.array(cluster_silscores + [silcoeff]),
index = centroids_names_red + ['Global'], columns = ['Sil. Score'] )

display(silcoeffs_df)

# visualization for avg silhouette scores
plt.figure(figsize=(10,6))
plt.bar(silcoeffs_df.index.tolist(), silcoeffs_df['Sil.
Score'].values)
plt.xlabel('data')
plt.ylabel('avg silhouette scores')
plt.title('INTERNAL EVALUATION - SILHOUETTE SCORES')
plt.grid(True, linestyle='--',linewidth=0.5,alpha=0.7)
plt.show
```

	Sil. Score
Tech-Action	0.179068
Introverted	0.161431
Emotional	0.177793
Entertainment	0.148107
Global	0.165676



Comment the results, also considering the results observed previously (e.g., score graphs, centroids, etc. - max 150 words):

Silhouette scores range from -1 to 1, where higher values indicate better separation and cohesion within clusters. The average scores for these clusters are all positive but relatively low, ranging from 0.148 to 0.179, suggesting that the clusters exist but are not sharply separated. The 'Tech-Action' and 'Emotional' clusters have the highest average scores, indicating stronger internal cohesion. These clusters also show a clearer distinction in gender composition, with 'Tech-Action' having more males and 'Emotional' more females relative to the global distribution. In contrast, the 'Entertainment' cluster has the lowest score and appears more heterogeneous. The global silhouette score (0.166) confirms moderate clustering quality. This is also reflected in the score graph, where the data points form a diffuse cloud, and cluster boundaries are not easily distinguishable.